科研写作入门指南

v1.1

前言

本教程为研究生提供一些科研写作入门指导。

参考 MIT CommKit articles（https://mitcommlab.mit.edu/find-comm-labs/#explorecommkit）等。

发布于。

注：蓝色为超链接或重要内容，红色为个人批注。

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目录

MIT-布罗德研究所

https://mitcommlab.mit.edu/broad/use-the-commkit/

论文写作

一般原则

如何撰写论文：

撰写论文时，从哪里开始取决于每个人。有些人喜欢直接写结果，有些人喜欢先写引言，还有些人喜欢先写方法。从你觉得最容易的开始，等到遇到瓶颈时再考虑是否需要改变策略。

如果你不知道从哪里开始，一般建议从结果开始。有证据表明，人们觉得这部分最容易写。而且，写完结果后，它们就像一个指南针，其他部分都以此为依据。

需要考虑的事项：

- 摘要：我如何简洁地传达我的结果的本质和意义？
- 简介：读者需要什么信息才能理解结果？
- 讨论：现在读者知道了结果，我可以向她提出哪些新的见解或问题？
- 方法：需要哪些方法来获得我的结果？

你的论文的主要思想或结论是什么？

虽然有效的写作风格有很多，但组织和结构对任何稿件都至关重要。以下是一些整理思路的方法：

1. 查看所有数据并将其细分为您执行的实验。
2. 写下每次实验得出的最重要的结论。
3. 确定这些结论之间的相互关系，以及如何对它们进行排序以形成合理的逻辑路径。(通常，这与您进行实验的顺序不同。)
4. 按照这个顺序列出实验的提纲。这种安排如何影响引言的结构？摘要的妙语？讨论部分提出的突破性问题？
5. 如果你觉得不确定，可以尝试重新安排实验顺序，然后重新开始。有时，多次重复此过程可能会发现应该包含一个额外的实验，或者某些结果不需要包含在正文中。

如果你仍然难以整理思路：

试着向同事或朋友进行简短的口头描述。这不需要花费太多时间，有时你会发现自己直观地了解了论文的逻辑流程。

- 熟悉你工作的同事可以帮助确定实验的逻辑路径
- 你的非科学家朋友或亲戚可以帮助你弄清楚如何清楚地描述论文的要点

如果这些方法不管用，那就开始写吧。即使提纲不完整，有时只要把文字写在纸上，就能让整个过程更加清晰。

组织技巧：

任何做过演讲、写过论文或基金申请书的人都经历过那种痛苦的挫败感：翻遍实验室笔记本，试图找到几个月（甚至几年）前实验的数据。回想起来，你会意识到，如果能把数据保存成随时可用的格式就好了，这样在正式汇报时，就能节省不少时间。以下是一些关于如何减轻数据整理和汇报压力的建议：

准备好你的数据图表：

- 记录实验细节。这听起来显而易见，但有时很容易忘记写下浓度单位或图表变量。随着时间的推移，这些信息可能不像你最初进行实验时那么明显。
- 及时进行“可发表”的实验。一旦获得“成功”实验的结果，请立即确认所有必要的重复实验、适当的对照实验和统计检验（如适用）均已完成。几个月后，如果再用更好的对照或更多的重复实验重新进行这些实验，可能会非常令人沮丧！
- 将结果格式化为图表。将所有数据保存为“图表格式”将加快写作速度，并确保你在最后一刻需要进行演示时做好准备。

整理参考文献：

- 考虑使用一些整理工具。Papers（https://www.papersapp.com/）或 ReadCube（https://www.readcube.com/）等应用程序都是不错的选择！（现在更推荐 EndNote 或 Noteexpress）
- 记录方法参考文献。如果你使用了需要参考的技术，请在实验旁边写上参考文献。几个月

/几年后再查找这些论文可能会很耗时。

- 保留一份可能有助于手稿引言或讨论部分的参考文献列表。

随着数据的发展，始终要求反馈：

- 向同事展示你的数据。他们不仅可以对你的实验提供反馈，还能让你知道你的数据是否清晰地呈现。
- 征求改进或补充实验的建议。有些实验室会召开小组会议和小组会议，以便持续获得反馈，但有些实验室则没有。在提交论文前最大限度地收集反馈有助于确保提交的是优化后的产品，并能帮助您预测提交后审阅者可能提出的要求。

写作之前

决定在何处发表是撰写论文的关键环节。这一决定将极大地影响论文的写作方式（面向哪些读者）、需要纳入哪些实验，以及稿件的整体格式（长度、图号、参考文献格式）。以下是开始撰写稿件前需要考虑的几点。

大多数期刊的网站上都有一个名为"作者指南 Author Guidelines"的版块。在这里，您可以找到向该期刊投稿所需的所有相关信息。找到这些信息后， 请列出期刊要求的投稿清单。

例如：

- 需要投稿信（cover or submission letter）吗？
- 该期刊是否提供稿件模板？（这可以大大加快格式化过程）
- 所需的引用格式是什么？
- 是否只接受某些类型的文件？（许多期刊指定首选或要求的文件类型和大小）
- 图形和文本是否需要特定的字体？
- 是否需要图形摘要？

在撰写稿件前，清晰地了解期刊的要求，可以最大程度地减少最后一刻重新排版、更改图表字体或文件大小的麻烦。这些改动看似微不足道，但累积起来可能会造成不必要的压力。

避免使用行话

避免使用专业术语可以让更广泛的科学受众理解你的研究成果，但同时你也不希望你的摘要过于高深，以至于专业读者感到无聊，并不清楚你具体做了什么。作为一种折衷方案，可以先对任何专业术语进行概括描述，然后再添加更专业的名称/细节（例如，与其说是“16S rRNA”，不如说是“细菌分类标记基因，16S rRNA”）。识别专业术语的一个好方法是请不同科学领域的同事或朋友通读你的摘要，并指出任何使你意思模糊的术语。

针对专家和新手读者调整你的写作风格

专家。您所在细分领域的一小群科学家，或称“专家”，会预先了解以下方面：1）您所在领域的知识现状；2）需要解答的主要问题；以及 3）常用的实验技术和分析方法。在专注于特定领域的期刊上发表文章时，您最有可能遇到这样的读者。为了使你的写作更贴近专家读者，请执行以下操作：

- 开门见山。先简短地介绍一下总体背景，然后在“具体背景”部分进一步明确。
- 包含更多细微差别。你可以在激励问题和方法中加入更多细节，这样你就能减少花费在帮助读者理解定义等方面的篇幅。

新手。通常，“新手”读者，或者说不属于你所在特定子领域的科学家，会在你的读者群中占据很大比例。通过让你的研究成果更容易被新手读者理解，你可以显著增加你的论文的阅读量。在这种情况下，引言将发挥极其重要的作用，让新手读者能够理解你的研究动机和研究成果。为了适应新手读者，你可以：

- 提供更多背景信息。在概括和具体背景方面，请略微退后几步。为了清晰起见，你可能会牺牲一些细节；你可能不会将背景缩小到“具体背景”那么具体，也可能会扩展整个背景部分。
- 避免使用专业术语。介绍/定义所有将在论文中讨论的“关键人物”和重要技术。
- 简化激励问题和结果。知识缺口中提出的问题越基础，人们就越会关注它。花点时间找出你的工作所回答的最简单、最根本的问题。

摘要

何时撰写摘要

1. 结果
2. 讨论
3. 引言
4. 摘要

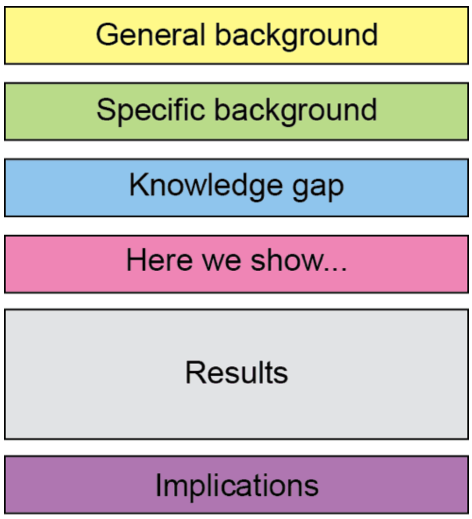
撰写摘要可能很困难，因为你的任务是将大量的工作浓缩到如此有限的篇幅中。为了方便起见，最好最后再写摘要。通读整篇论文，并将每个部分提炼出其要点。有时，通过减法来回答这个问题会很有帮助。例如，如果你想要提炼你的研究结果，只需列出你所有的发现，然后逐一查看列表，逐一划掉或合并每个发现，直到只剩下最重要的结果。

目的

标题和摘要是感兴趣的读者在浩如烟海的科学出版物、海报或会议演讲中找到您研究成果的主要媒介。当同行科学家偶然看到您的摘要时，他们会快速浏览，看看是否值得花时间深入研究论文正文。因此，摘要的主要目的是以简洁易懂的方式阐述和描述您的研究工作。这将确保您的目标受众能够找到并阅读您的科研成果。

摘要公式

通过以可预测的顺序提供信息来实现清晰度：成功的摘要由 6 个有序部分组成，称为“摘要公式”。



一般背景和具体背景（每句约 1 句）。介绍你将要讨论的科学领域及其知识现状。先从一般背景开始，然后逐步缩小到与论文主题相关的主题。如果你使用专业术语，请务必对其进行简要定义。

知识差距（约 1 句话）。既然你已经陈述了已知的内容，那就说明未知的内容。你的研究试图回答什么具体问题？

“我们在这里展示……”（约 1 句话）。请阐述你的一般实验方法，并回答你刚才在“知识差距”部分提出的问题。

实验方法与结果（约 1-3 句）。请详细描述您最重要的方法和结果。您是如何得出“我们在此展示……”部分中提出的结论的？

启示（约 1 句话）。 描述你的研究结果如何影响我们对相关领域的理解，以及/或对未来研究的影响。

带注释的示例：

LETTER

doi:10.1038/nature10571

Ecology drives a global network of gene exchange connecting the human microbiome

Chris S. Smillie[1]*, Mark B. Smith[2]*, Jonathan Friedman[1], Otto X. Cordero[3], Lawrence A. David[4] & Eric J. Alm[3,5,6]

general background

specific background

knowledge gap

here we show...

results with key, concrete values

meaning of results

Horizontal gene transfer (HGT), the acquisition of genetic material from non-parental lineages, is known to be important in bacterial evolution[1,2]. In particular, HGT provides rapid access to genetic innovations, allowing traits such as virulence[3], antibiotic resistance[4] and xenobiotic metabolism[5] to spread through the human microbiome. Recent anecdotal studies providing snapshots of active gene flow on the human body have highlighted the need to determine the frequency of such recent transfers and the forces that govern these events[4,5]. Here we report the discovery and characterization of a vast, human-associated network of gene exchange, large enough to directly compare the principal forces shaping HGT. We show that this network of 10,770 unique, recently transferred (more than 99% nucleotide identity) genes found in 2,235 full bacterial genomes, is shaped principally by ecology rather than geography or phylogeny, with most gene exchange occurring between isolates from ecologically similar, but geographically separated, environments. For example, we observe 25-fold more HGT between human-associated bacteria than among ecologically diverse non-human isolates ($P = 3.0 \times 10^{-270}$). We show that within the human microbiome this ecological architecture continues across multiple spatial scales, functional classes and ecological niches with transfer further enriched among bacteria that inhabit the same body site, have the same oxygen tolerance or have the same ability to cause disease. This structure offers a window into the molecular traits that define ecological niches, insight that we use to uncover sources of antibiotic resistance and identify genes associated with the pathology of meningitis and other diseases.

The human body is a complex biological network comprising ten

acquired from close relatives, because these genes have greater compatibility with native molecular machinery[15,16].

Geography might provide an alternative structure to HGT by restricting dispersal, as suggested by the geographically organized distribution of *Vibrio cholera* integrons[17] and NDM-1 antibiotic resistance genes[18].

A third possibility is that ecological similarity shapes networks of gene exchange by selecting for the transfer and proliferation of adaptive traits or by increasing physical interactions between community members. Reports of enriched levels of HGT between hyperthermophiles[19] and spatially segregated exchange among *Shewanella* isolates[20] offer suggestive glimpses of such an ecological structure. However, it has been difficult to determine whether ecology has a broader function in HGT because of the limited availability of genomes from similar environments and because most previous work has ignored the distinction between recent transfers and ancient events. The inclusion of transfers from millions or billions of years in the past can obscure ecological structure, because historical niches may not reflect modern environmental associations.

To explore the effects of phylogeny, geography and ecology on HGT we use an evolutionary-rate heuristic to identify recent transfers between thousands of microbial genomes. Our heuristic finds blocks of nearly identical DNA (more than 500 nucleotides, more than 99% identity) in distantly related genomes (less than 97% 16S rRNA similarity). HGT is the best explanation for these observations because the highly conserved 16S gene evolves about 25-fold more slowly than protein-coding synonymous sites[21]. As a result, vertically inherited orthologues in such divergent genomes are nearly saturated with mutations at

ARTICLE

doi:10.1038/nature13824

Clonal dynamics of native haematopoiesis

Jianlong Sun[1,2,3], Azucena Ramos[1], Brad Chapman[4], Jonathan B. Johnnidis[5], Linda Le[1], Yu-Jui Ho[6], Allon Klein[7], Oliver Hofmann[4] & Fernando D. Camargo[1,2,3]

general background

specific background

here we show...

knowledge gap

It is currently thought that life-long blood cell production is driven by the action of a small number of multipotent haematopoietic stem cells. Evidence supporting this view has been largely acquired through the use of functional assays involving transplantation. However, whether these mechanisms also govern native non-transplant haematopoiesis is entirely unclear. Here we have established a novel experimental model in mice where cells can be uniquely and genetically labelled *in situ* to address this question. Using this approach, we have performed longitudinal analyses of clonal dynamics in adult mice that reveal unprecedented features of native haematopoiesis. In contrast to what occurs following transplantation, steady-state blood production is maintained by the successive recruitment of thousands of clones, each with a minimal contribution to mature progeny. Our results demonstrate that a large number of long-lived progenitors, rather than classically defined haematopoietic stem cells, are the main drivers of steady-state haematopoiesis during most of adulthood. Our results also have implications for understanding the cellular origin of haematopoietic disease.

results, including the methodological approach and the interpretation of the results

implication of results (they even say "implications"!)

引言

何时撰写引言

1. 结果
2. 讨论
3. 引言
4. 摘要

目的

论文引言是向读者提供理解论文所需背景信息的机会：您所在领域的知识现状、促使您开展研究的问题及其意义、您如何尝试回答该问题（方法）以及您的主要发现。一篇写得好的引言可以让更多读者理解您的研究成果，从而扩大您的读者群。

引言公式

清晰的表达是通过按可预测的顺序提供信息来实现的。因此，成功的引言由四个有序的要素组成，这四个要素被称为“引言公式”。

1. 一般背景。介绍你的项目所涉及的科学领域，并强调我们对该系统的理解现状。
2. 具体背景。缩小论文要讨论的子领域，并再次强调我们对该领域的理解程度。

   提示：只需向读者提供理解系统所需的技术细节，无需赘述。你的目的并非展示你的知识广度，而是为读者提供理解你的结果及其意义所需的所有工具。
3. 知识差距。（即提出问题）在讨论了我们已知的知识之后，阐明我们未知的知识，并特别关注激发你进行研究的问题。前两个部分应该为这个问题做铺垫。也就是说，基于你在一般和具体背景中所描述的内容，激发你进行研究的问题应该是合乎逻辑的下一步。
4. “我们在此展示……”非常简要地概括您的研究方法和发现。请注意，您可以在本节结尾

处用一两句话阐述您的研究结果的意义/创新性，但这并非必需，因为您将在讨论部分更详细地讨论这些要点。

带注释的示例：

Identifying Gut Microbe–Host Phenotype Relationships Using Combinatorial Communities in Gnotobiotic Mice
Faith *et al.*, *Science Translational Medicine* (2014) doi: 10.1126/scitranslmed.3008051

microbiome as a basis for disease is a good starting point for almost anyone (and esp. for translational medicine folks!)

general background

Characterizations of the structural configurations of human gut microbial communities are beginning to reveal differences between healthy individuals and those with various diseases (1–6). Experiments involving transplantation of intact uncultured microbiota from healthy humans to humans with colitis induced by *Clostridium difficile* or patients with metabolic syndrome have helped to establish a causal role for the microbiota in these disorders and, at the same time, have provided proof of principle that the microbiota represents a therapeutic target for treating or preventing disease (6, 7).

specific background

nitty gritty: how could the microbiome be used as treatment for disease?

Transplantation of intact uncultured human gut microbiota samples from human donors with various physiologic or disease states, or cultured members of the microbiota, to germ-free mice provides an opportunity to identify specific microbial species that may influence the physiologic, metabolic, and immunologic properties of humans (8–12). One challenge has been to develop a scalable, unbiased approach for identifying human gut bacterial strains that modulate phenotypic variation in recipient mice. Here, we describe such a method (Fig. 1). It began with a screen of gnotobiotic mice containing transplanted intact uncultured fecal microbiota from different human donors to identify transmissible phenotypes that could be attributed to each donor's microbiota. We then generated a clonally arrayed collection of cultured anaerobic bacteria in multiwell plates from a donor whose intact uncultured microbiota transmitted a phenotype of interest. Each well of the plate harbored a bacterial strain whose genome had been sequenced (13, 14). The arrayed culture collection of bacteria was then randomly fractionated into subsets of various sizes. Each subset was gavaged into a germ-free animal, individually maintained in a sterile filter-topped cage, to observe the effect of the bacterial consortium on a host phenotype. By repeating this process across many subsets, the effect of each strain in the arrayed culture collection was assayed in the context of a diverse background of community memberships and sizes. Feature selection algorithms and follow-up experiments where mice were colonized with single strains (monocolonization) were then used to identify the strains whose presence or absence best explained the observed phenotypic variation. Using this approach, we have identified bacterial strains that modulate adiposity, intestinal metabolite composition, and the immune system.

a short, sweet, but fairly narrow knowledge gap

the items in the "here we show" let you know what the Results will look like

Article

Cell

Cpf1 Is a Single RNA-Guided Endonuclease of a Class 2 CRISPR-Cas System

Bernd Zetsche,[1,2,3,4,5,10] Jonathan S. Gootenberg,[1,2,3,4,6,10] Omar O. Abudayyeh,[1,2,3,4] Ian M. Slaymaker,[1,2,3,4] Kira S. Makarova,[7] Patrick Essletzbichler,[1,2,3,4] Sara E. Volz,[1,2,3,4] Julia Joung,[1,2,3,4] John van der Oost,[8] Aviv Regev,[1,9] Eugene V. Koonin,[7] and Feng Zhang[1,2,3,4,*]
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SUMMARY

The microbial adaptive immune system CRISPR mediates defense against foreign genetic elements through two classes of RNA-guided nuclease effectors. Class 1 effectors utilize multi-protein complexes, whereas class 2 effectors rely on single-component effector proteins such as the well-characterized Cas9. Here, we report characterization of Cpf1, a putative class 2 CRISPR effector. We demonstrate that Cpf1 mediates robust DNA interference with features distinct from Cas9. Cpf1 is a single RNA-guided endonuclease lacking tracrRNA, and it utilizes a T-rich protospacer-adjacent motif. Moreover, Cpf1 cleaves DNA via a staggered DNA double-stranded break. Out of 16 Cpf1-family proteins, we identified two candidate enzymes from *Acidaminococcus* and *Lachnospiraceae*, with efficient genome-editing activity in human cells. Identifying this mechanism of interference broadens our understanding of CRISPR-Cas systems and advances their genome editing applications.

INTRODUCTION

general background

Almost all archaea and many bacteria achieve adaptive immunity through a diverse set of CRISPR-Cas (clustered regularly interspaced short palindromic repeats and CRISPR-associated proteins) systems, each of which consists of a combination of Cas effector proteins and CRISPR RNAs (crRNAs) (Makarova et al., 2011, 2015). The defense activity of the CRISPR-Cas systems includes three stages: (1) adaptation, when a complex of Cas proteins excises a segment of the target DNA (known as a protospacer) and inserts it into the CRISPR array (where this sequence becomes a spacer); (2) expression and processing of the precursor CRISPR (pre-cr) RNA resulting in the formation of mature crRNAs; and (3) interference, when the effector module—either another Cas protein complex or a single large protein—is guided by a crRNA to recognize and cleave target DNA (or in some cases, RNA) (Horvath and Barrangou, 2010; Sorek et al., 2013; Barrangou and Marraffini, 2014). The adaptation stage is mediated by the complex of the Cas1 and Cas2 proteins, which are shared by all known CRISPR-Cas systems, and sometimes involves additional Cas proteins. Diversity is observed at the level of processing of the pre-crRNA to mature crRNA guides, proceeding via either a Cas6-related ribonuclease or a housekeeping RNaseIII that specifically cleaves double-stranded RNA hybrids of pre-crRNA and tracrRNA. Moreover, the effector modules differ substantially among the CRISPR-Cas systems (Makarova et al., 2011, 2015; Charpentier et al., 2015). In the latest classification, the diverse CRISPR-Cas systems are divided into two classes according to the configuration of their effector modules: class 1 CRISPR systems utilize several Cas proteins and the crRNA to form an effector complex, whereas class 2 CRISPR systems employ a large single-component Cas protein in conjunction with crRNAs to mediate interference (Makarova et al., 2015).

specific background

(with all acronyms defined, with classes of CRISPR introduced, …)

Multiple class 1 CRISPR-Cas systems, which include the type I and type III systems, have been identified and functionally characterized in detail, revealing the complex architecture and dynamics of the effector complexes (Brouns et al., 2008; Marraffini and Sontheimer, 2008; Hale et al., 2009; Sinkunas et al., 2013; Jackson et al., 2014; Mulepati et al., 2014). Several class 2 CRISPR-Cas systems have also been identified and experimentally characterized, but they are all type II and employ homologous RNA-guided endonucleases of the Cas9 family as effectors (Barrangou et al., 2007; Garneau et al., 2010; Deltcheva et al., 2011; Sapranauskas et al., 2011; Jinek et al., 2012; Gasiunas et al., 2012). A second, putative class 2 CRISPR system, tentatively assigned to type V, has

knowledge gap

been recently identified in several bacterial genomes (http://www.jcvi.org/cgi-bin/tigrfams/HmmReportPage.cgi?acc=TIGR04330) (Schunder et al., 2013; Vestergaard et al., 2014; Makarova et al., 2015). The putative type V CRISPR-Cas systems contain a large, ~1,300 amino acid protein called Cpf1 (CRISPR from *Prevotella* and *Francisella* 1). It remains unknown, however, whether Cpf1-containing CRISPR loci indeed represent functional CRISPR systems. Given the broad applications of Cas9 as a genome-engineering tool (Hsu et al., 2014; Jiang and Marraffini, 2015), we sought to explore the function of Cpf1-based putative CRISPR systems.

why reader should care

Here, we show that Cpf1-containing CRISPR-Cas loci of *Francisella novicida U112* encode functional defense systems capable of mediating plasmid interference in bacterial cells guided by the CRISPR spacers. Unlike Cas9 systems, Cpf1-containing CRISPR systems have three features. First, Cpf1-associated CRISPR arrays are processed into mature crRNAs without the requirement of an additional *trans*-activating crRNA (tracrRNA) (Deltcheva et al., 2011; Chylinski et al., 2013). Second, Cpf1-crRNA complexes efficiently cleave target DNA proceeded by a short T-rich protospacer-adjacent motif (PAM), in contrast to the G-rich PAM following the target DNA for Cas9 systems. Third, Cpf1 introduces a staggered DNA double-stranded break with a 4 or 5-nt 5′ overhang.

here we show…

To explore the suitability of Cpf1 for genome-editing applications, we characterized the RNA-guided DNA-targeting requirements for 16 Cpf1-family proteins from diverse bacteria, and we identified two Cpf1 enzymes from *Acidaminococcus sp. BV3L6* and *Lachnospiraceae bacterium ND2006* that are capable of mediating robust genome editing in human cells. Collectively, these results establish Cpf1 as a class 2 CRISPR-Cas system that includes an effective single RNA-guided endonuclease with distinct properties that has the potential to substantially advance our ability to manipulate eukaryotic genomes.

… with preview of results

RESULTS

Cpf1-Containing CRISPR Loci Are Active Bacterial Immune Systems

Cpf1 was first annotated as a CRISPR-associated gene in TIGRFAM (http://www.jcvi.org/cgi-bin/tigrfams/HmmReportPage.cgi?acc=TIGR04330) and has been hypothesized to be the effector of a CRISPR locus that is distinct from the Cas9-containing type II CRISPR-Cas loci that are also present in the genomes of some of the same bacteria, such as multiple strains of *Francisella* and *Prevotella* (Schunder et al., 2013; Vestergaard et al., 2014; Makarova et al., 2015) (Figure 1A). The Cpf1 protein contains a predicted RuvC-like endonuclease domain that is distantly related to the respective nuclease domain of Cas9. However, Cpf1 differs from Cas9 in that it lacks a second, HNH endonuclease domain, which is inserted within the

方法

成功标准

成功的方法部分：

- 提供选择方法的理由
- 允许读者通过复制来确认你的发现

结构

比较方法和结果的真实注释示例。注意两部分之间副标题的对应性。

确定你的目的

方法部分的目的是描述引言中提出的问题/知识缺口是如何在结果部分得到解答的。并非所有读者都会对此信息感兴趣。对于那些感兴趣的读者来说，方法部分有两个目的：

1. 让读者判断研究结果和结论是否有效。

   对结果的解读取决于您获取结果的方法。对您的结果持怀疑态度的读者会阅读“方法”部分，以判断其可信度。他们希望了解您是否选择了最合适的方法并进行了必要的对照。如果没有这些内容，持怀疑态度的读者可能会认为您的数据以及由此得出的任何结论都不可靠。

2. 允许读者重复学习。

   对于有兴趣复制您的研究的读者，“方法”部分应该提供足够的信息，以便他们获得相同或相似的结果。

分析你的受众

通常，只有您所在领域的读者才会想要重复您的研究，或者具备评估您的方法论的知识。更普通的读者会阅读引言，然后直接阅读结果部分。因此，您可以假设阅读您研究方法的人了解您所在领域常用的方法论。为了评估特定方法所需的详细程度，您可以查阅目标期刊上先前发表的文章。

如果你的论文旨在吸引多个领域的专家，你仍然需要为一组专家撰写方法。例如，假设你应用了一种新颖的计算方法来对一个已充分表征的生物系统获得新的见解。你的目标是向生物学家展示你的计算工具的价值，还是向计算科学家展示它们如何帮助研究生物学？在前一

种情况下，假设计算专业知识较少：提供更详尽的解释来解释方法的工作原理以及选择它们的原因。

技能

说明选择方法的原因

想要评估您的方法论的读者会阅读“方法”部分来评判您的实验设计。在描述您的方法时，应重点强调您为何应用该方法，而不是如何执行该方法。例如，您无需解释如何进行蛋白质印迹实验，但您可能需要描述为什么蛋白质印迹实验适合当前任务（并且，可能还需要说明为什么您没有使用其他方法）。

明确检测目的	“为了验证蛋白质产物是否存在，我们进行了蛋白质印迹实验。”
解释为什么使用特定方法	“为了最大限度地增加候选基因的数量，我们使用了统计方法 X，该方法优先减少假阴性错误，而不是减少假阳性。”
解释为什么你没有使用其他方法	“对该生物体基因组中一小部分有用的片段进行了测序。对整个基因组进行测序可以提供更多信息，但成本高昂。”

使用副标题来组织内容

建议在“结果”部分使用“方法”中的副标题来对相关实验进行分组，并建立逻辑流程。先撰写“结果”部分，然后按照“结果”副标题的顺序撰写“方法”。平行结构有助于读者轻松找到两部分中的对应信息。

方法和结果的副标题可能不完全对应。有时，您可能需要多个“方法”副标题来解释一个“结果”副标题。有时，一个“方法”副标题足以解释多个“结果”副标题。

提供最少的必要细节

仅提供读者能够重复研究中提出的实验所需的细节；过多的细节都是多余的。请记住，读者会使用方法来评估你结论的有效性，因此请详细说明任何可能导致读者得出不同结论的方法学细节。

您可以引用标准方法的论文，但任何修改或变更都应清晰说明。引用方法时，请引用描述该方法的原始论文，而不是使用该方法的论文。（此处说明了引用源头文献的重要性）这有助

于避免读者必须循着一连串的引用才能找到有关该方法的信息。

避免使用“we did…”或“the authors did…”

方法部分应重点关注实验，而不是作者。避免使用“We/The authors did...”这样的表述方式，即使这需要你使用被动语态。

“Samples were processed with standard DNA extraction protocols.”

而非

“We processed the samples with standard DNA extraction protocols.”

带注释的示例：

Generation of Cpf1 Protein Lysate

Cpf1 proteins codon optimized for human expression were synthesized with a C-terminal nuclear localization tag and cloned into the pcDNA3.1 expression plasmid by Genscript (Table S1). 2,000 ng of Cpf1 expression plasmids were transfected into 6-well plates of HEK293FT cells at 90% confluency using Lipofectamine 2000 reagent (Life Technologies). 48 hr later, cells were harvested by washing once with DPBS (Life Technologies) and scraping in lysis buffer (20 mM HEPES [pH 7.5], 100 mM KCl, 5mM $MgCl_2$, 1 mM DTT, 5% glycerol, 0.1% Triton X-100, 1X cOmplete Protease Inhibitor Cocktail Tablets [Roche]). Lysate was sonicated for 10 min in a Biorupter sonicator (Diagenode) and then centrifuged. Supernatant was frozen for subsequent use in in vitro cleavage assays.

In Vitro Cleavage Assay

Cleavage in vitro was performed either with purified protein (25 nM) or mammalian lysate with protein at 37 °C in cleavage buffer (NEBuffer 3, 5 mM DTT) for 20 min. The cleavage reaction used 500 ng of synthesized crRNA or sgRNA and 200 ng of target DNA. Target DNA involved either protospacers cloned into pUC19 or PCR amplicons of gene regions from genomic DNA isolated from HEK293 cells. Reactions were cleaned up using PCR purification columns (QIAGEN) and were run on 2% agarose E-gels (Life Technologies). For native and denaturing gels to analyze cleavage by nuclease mutants, cleaned-up reactions were run on TBE 6% polyacrylamide or TBE-Urea 6% polyacrylamide gels (Life Technologies).

In Vitro Cpf1-Family Protein PAM Screen

In vitro cleavage reactions with Cpf1-family proteins were run on 2% agarose E-gels (Life Technologies). Bands corresponding to un-cleaved target were gel extracted using QIAquick Gel Extraction Kit (QIAGEN), and the target PAM region was amplified and sequenced using a MiSeq (Illumina) with a single-end 150 cycle kit. Sequencing results were entered into the PAM discovery pipeline.

Western Blot Analysis

Cells were lysed in 1 × RIPA buffer (Cell Signaling Technology) supplemented with protease inhibitor cocktail (Roche). Equal volumes of cell lysate were run on BOLT 4%–12% Bis-Tris gradient gels (Invitrogen) and transferred to PVDF membranes (Millipore). Non-specific antigen binding was blocked with TBS-T (50 mM Tris, 150 mM NaCl and 0.05% Tween-20) with 5% BLOT-QuickBlocker Reagent (Millipore) for 1 hr. Membranes were incubated with primary antibodies (anti-HA-tag [Cell Signaling Technology C29F4] or HRP-conjugated GAPDH [Cell Signaling Technology 14C10]) for 1 hr in TBS-T with 1% BLOT-QuickBlocker. Membranes were washed for three 10 min washes and anti-HA-tag membranes were further incubated with anti-rabbit antibody (Cell Signaling Technology 7074) for 1 hr followed by six 10 min washes in TBS-T. Proteins were visualized with West Pico Chemiluminescent Substrate (Life Technology) and imaged using the ChemiDoc MP Imaging System (Bio-Rad) and processed with ImageLab software (Bio-Rad).

SURVEYOR Nuclease Assay for Genome Modification

PCR amplicons comprised of a U6 promoter driving expression of the crRNA sequence were generated using Herculase II (Agilent Technologies) and appropriate U6 reverse primers (Table S2). 400 ng of Cpf1 expression plasmids and 100 ng of the U6::crRNA expression cassettes were transfected into 24-well plates of HEK293FT cells at 75%–90% confluency using Lipofectamine 2000 (Life Technologies).

Cells were incubated at 37 °C for 72 hr post-transfection before genomic DNA extraction. Genomic DNA was extracted using the QuickExtract DNA Extraction Solution (Epicenter) following the manufacturer's protocol. The genomic region flanking the CRISPR target site for each gene was PCR amplified, and products were purified using QiaQuick Spin Column (QIAGEN) following the manufacturer's protocol. 200–500 ng total of the purified PCR products were mixed with 1 μl 10 × Taq DNA Polymerase PCR buffer (Enzymatics) and ultrapure water to a final volume of 10 μl and were subjected to a re-annealing process to enable heteroduplex formation: 95 °C for 10 min, 95°C to 85 °C ramping at −2°C/s, 85 °C to 25 °C at −0.25 °C/s, and 25 °C hold for 1 min. After re-annealing, products were treated with SURVEYOR nuclease and SURVEYOR enhancer S (Integrated DNA Technologies) following the manufacturer's recommended protocol and analyzed on 4%–20% Novex TBE polyacrylamide gels (Life Technologies). Gels were stained with SYBR Gold DNA stain (Life Technologies) for 10 min and imaged with a Gel Doc gel imaging system (Bio-rad). Quantification was based on relative band intensities. Indel percentage was determined by the formula, $100 \times (1 - \mathrm{sqrt}(1 - (b + c)/(a + b + c)))$, where a is the integrated intensity of the undigested PCR product, and b and c are the integrated intensities of each cleavage product.

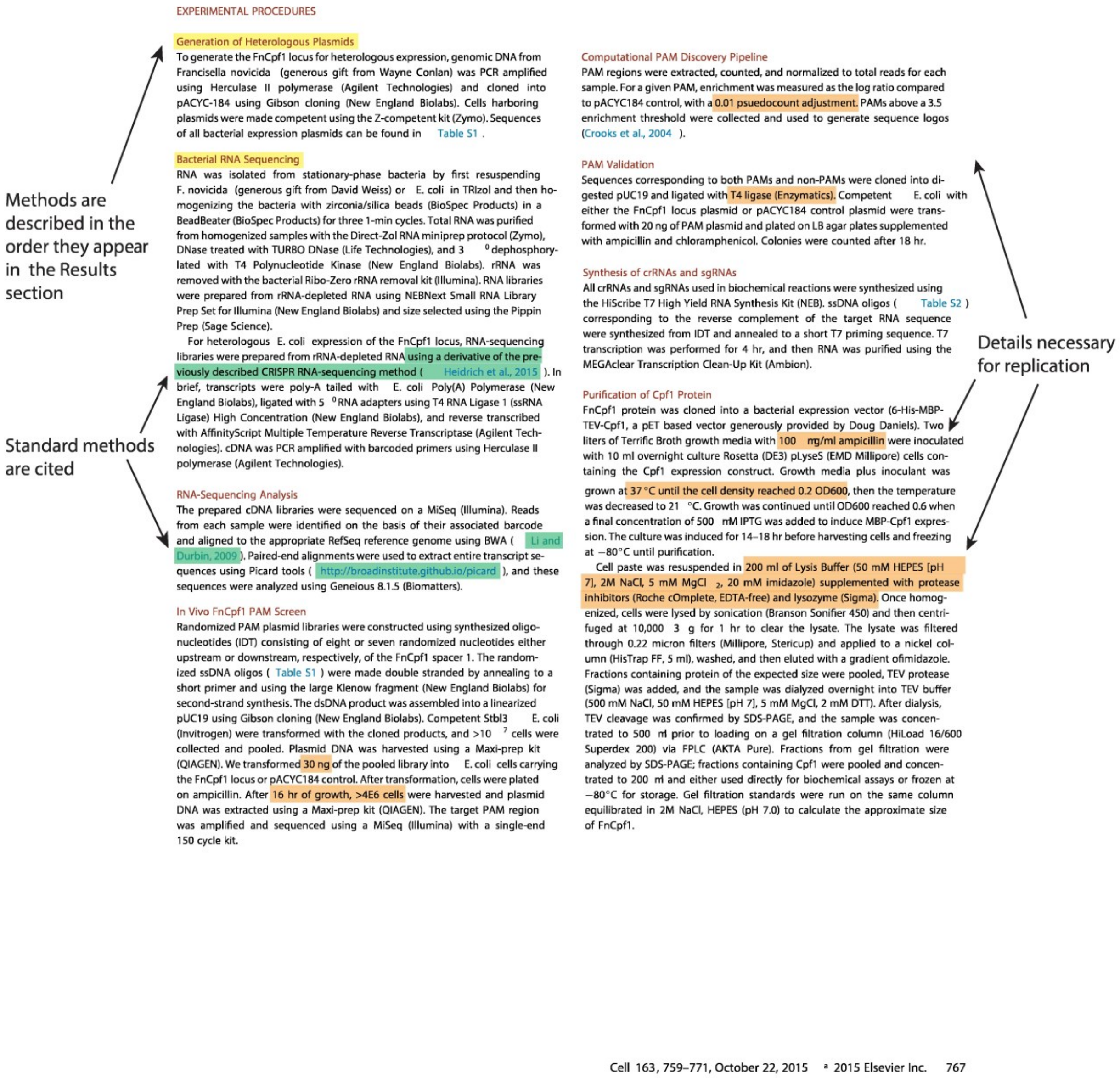
Deep Sequencing to Characterize Cpf1 Indel Patterns in 293FT Cells

HEK293FT cells were transfected and harvested as described for assessing activity of Cpf1 cleavage. The genomic-region-flanking DNMT1 targets were amplified using a two-round PCR region to add Illumina P5 adapters as well as unique sample-specific barcodes to the target amplicons. PCR products were run on 2% E-gel (Invitrogen) and gel extracted using QiaQuick Spin Column (QIAGEN) as per the manufacturer's recommended protocol. Samples were pooled and quantified by Qubit 2.0 Fluorometer (Life Technologies). The prepared cDNA libraries were sequenced on a MiSeq with a single-end 300 cycle kit (Illumina). Indels were mapped using a Python implementation of the Geneious 6.0.3 Read Mapper.

Reasons for using a particular methodology

Computational Analysis of Cpf1 loci

PSI-BLAST program (Altschul et al., 1997) was used to identify Cpf1 homologs in the NCBI NR database using several known Cpf1 sequences as queries with the Cpf1 with the E-value cut-off of 0.01 and low-complexity filtering and composition-based statistics turned off. The TBLASTN program with the E-value cut-off of 0.01 and low-complexity filtering turned off was used to search the NCBI WGS database using the Cpf1 profile (Makarova et al., 2015) as the query. Results of all searches were combined (Table S3). The HHpred program was used with default parameters (Söding et al., 2006) to identify remote sequence similarity using a subset of representative Cpf1 sequences queries. Multiple sequence alignments were constructed using MUSCLE (Edgar, 2004) with manual correction based on pairwise alignments obtained using PSI-BLAST and HHpred programs. Phylogenetic analysis was performed using the FastTree program with the WAG evolutionary model and the discrete gamma model with 20 rate categories (Price et al., 2010). Protein secondary structure was predicted using Jpred 4 (Drozdetskiy et al., 2015). CRISPR repeats were identified using PILER-CR (Edgar, 2007) and CRISPRfinder (Grissa et al., 2007).



EXPERIMENTAL PROCEDURES

Generation of Heterologous Plasmids

To generate the FnCpf1 locus for heterologous expression, genomic DNA from Francisella novicida (generous gift from Wayne Conlan) was PCR amplified using Herculase II polymerase (Agilent Technologies) and cloned into pACYC-184 using Gibson cloning (New England Biolabs). Cells harboring plasmids were made competent using the Z-competent kit (Zymo). Sequences of all bacterial expression plasmids can be found in Table S1 .

Bacterial RNA Sequencing

RNA was isolated from stationary-phase bacteria by first resuspending F. novicida (generous gift from David Weiss) or E. coli in TRIzol and then homogenizing the bacteria with zirconia/silica beads (BioSpec Products) in a BeadBeater (BioSpec Products) for three 1-min cycles. Total RNA was purified from homogenized samples with the Direct-Zol RNA miniprep protocol (Zymo), DNase treated with TURBO DNase (Life Technologies), and 3 ^{0} dephosphorylated with T4 Polynucleotide Kinase (New England Biolabs). rRNA was removed with the bacterial Ribo-Zero rRNA removal kit (Illumina). RNA libraries were prepared from rRNA-depleted RNA using NEBNext Small RNA Library Prep Set for Illumina (New England Biolabs) and size selected using the Pippin Prep (Sage Science).

For heterologous E. coli expression of the FnCpf1 locus, RNA-sequencing libraries were prepared from rRNA-depleted RNA using a derivative of the previously described CRISPR RNA-sequencing method (Heidrich et al., 2015). In brief, transcripts were poly-A tailed with E. coli Poly(A) Polymerase (New England Biolabs), ligated with 5 ^{0} RNA adapters using T4 RNA Ligase 1 (ssRNA Ligase) High Concentration (New England Biolabs), and reverse transcribed with AffinityScript Multiple Temperature Reverse Transcriptase (Agilent Technologies). cDNA was PCR amplified with barcoded primers using Herculase II polymerase (Agilent Technologies).

RNA-Sequencing Analysis

The prepared cDNA libraries were sequenced on a MiSeq (Illumina). Reads from each sample were identified on the basis of their associated barcode and aligned to the appropriate RefSeq reference genome using BWA (Li and Durbin, 2009). Paired-end alignments were used to extract entire transcript sequences using Picard tools (http://broadinstitute.github.io/picard), and these sequences were analyzed using Geneious 8.1.5 (Biomatters).

In Vivo FnCpf1 PAM Screen

Randomized PAM plasmid libraries were constructed using synthesized oligonucleotides (IDT) consisting of eight or seven randomized nucleotides either upstream or downstream, respectively, of the FnCpf1 spacer 1. The randomized ssDNA oligos (Table S1) were made double stranded by annealing to a short primer and using the large Klenow fragment (New England Biolabs) for second-strand synthesis. The dsDNA product was assembled into a linearized pUC19 using Gibson cloning (New England Biolabs). Competent Stbl3 E. coli (Invitrogen) were transformed with the cloned products, and >10 ^{7} cells were collected and pooled. Plasmid DNA was harvested using a Maxi-prep kit (QIAGEN). We transformed 30 ng of the pooled library into E. coli cells carrying the FnCpf1 locus or pACYC184 control. After transformation, cells were plated on ampicillin. After 16 hr of growth, >4E6 cells were harvested and plasmid DNA was extracted using a Maxi-prep kit (QIAGEN). The target PAM region was amplified and sequenced using a MiSeq (Illumina) with a single-end 150 cycle kit.

Computational PAM Discovery Pipeline

PAM regions were extracted, counted, and normalized to total reads for each sample. For a given PAM, enrichment was measured as the log ratio compared to pACYC184 control, with a 0.01 psuedocount adjustment. PAMs above a 3.5 enrichment threshold were collected and used to generate sequence logos (Crooks et al., 2004).

PAM Validation

Sequences corresponding to both PAMs and non-PAMs were cloned into digested pUC19 and ligated with T4 ligase (Enzymatics). Competent E. coli with either the FnCpf1 locus plasmid or pACYC184 control plasmid were transformed with 20 ng of PAM plasmid and plated on LB agar plates supplemented with ampicillin and chloramphenicol. Colonies were counted after 18 hr.

Synthesis of crRNAs and sgRNAs

All crRNAs and sgRNAs used in biochemical reactions were synthesized using the HiScribe T7 High Yield RNA Synthesis Kit (NEB). ssDNA oligos (Table S2) corresponding to the reverse complement of the target RNA sequence were synthesized from IDT and annealed to a short T7 priming sequence. T7 transcription was performed for 4 hr, and then RNA was purified using the MEGAclear Transcription Clean-Up Kit (Ambion).

Purification of Cpf1 Protein

FnCpf1 protein was cloned into a bacterial expression vector (6-His-MBP-TEV-Cpf1, a pET based vector generously provided by Doug Daniels). Two liters of Terrific Broth growth media with 100 mg/ml ampicillin were inoculated with 10 ml overnight culture Rosetta (DE3) pLyseS (EMD Millipore) cells containing the Cpf1 expression construct. Growth media plus inoculant was grown at 37 °C until the cell density reached 0.2 OD600, then the temperature was decreased to 21 °C. Growth was continued until OD600 reached 0.6 when a final concentration of 500 mM IPTG was added to induce MBP-Cpf1 expression. The culture was induced for 14–18 hr before harvesting cells and freezing at −80°C until purification.

Cell paste was resuspended in 200 ml of Lysis Buffer (50 mM HEPES [pH 7], 2M NaCl, 5 mM $MgCl_2$, 20 mM imidazole) supplemented with protease inhibitors (Roche cOmplete, EDTA-free) and lysozyme (Sigma). Once homogenized, cells were lysed by sonication (Branson Sonifier 450) and then centrifuged at 10,000 3 g for 1 hr to clear the lysate. The lysate was filtered through 0.22 micron filters (Millipore, Stericup) and applied to a nickel column (HisTrap FF, 5 ml), washed, and then eluted with a gradient ofimidazole. Fractions containing protein of the expected size were pooled, TEV protease (Sigma) was added, and the sample was dialyzed overnight into TEV buffer (500 mM NaCl, 50 mM HEPES [pH 7], 5 mM MgCl, 2 mM DTT). After dialysis, TEV cleavage was confirmed by SDS-PAGE, and the sample was concentrated to 500 ml prior to loading on a gel filtration column (HiLoad 16/600 Superdex 200) via FPLC (AKTA Pure). Fractions from gel filtration were analyzed by SDS-PAGE; fractions containing Cpf1 were pooled and concentrated to 200 ml and either used directly for biochemical assays or frozen at −80°C for storage. Gel filtration standards were run on the same column equilibrated in 2M NaCl, HEPES (pH 7.0) to calculate the approximate size of FnCpf1.

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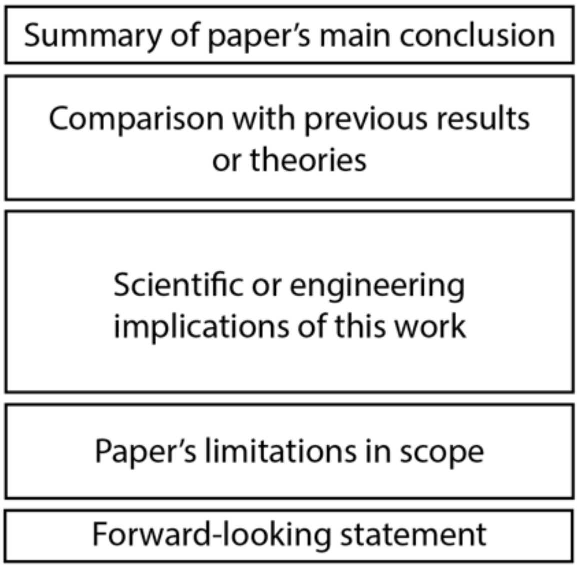
讨论

成功标准

强大的讨论部分：

1. 用一两句话讲述论文的主要结论。
2. 说明论文的结果如何有助于回答引言中提出的重大问题。
3. 解释这项工作如何（以及为什么）与其他类似工作一致或不一致。
4. 解释这项研究的局限性如何导致重大问题未得到解答。
5. 说明本文结果的扩展将如何有助于回答重大问题。

结构图



目的

讨论是论文的一部分，你可以在这里分享你认为你的研究结果对你在引言中提出的重大问题的意义。引言和讨论是天然的搭档：引言告诉读者你正在研究什么问题以及你为什么要做这个实验来探究它；讨论则告诉读者该实验的结果对更大的问题有何启示。

想象一下，你向一位实验室同事解释论文中的结果，他看起来很困惑，并问你："当然，但那又怎样？这为什么很酷或有趣？" 你对实验室同事的回答应该与讨论中的内容类似。

分析你的受众

不同类型的读者对你的讨论有不同的期望。非你所在领域的专家可能会先阅读你的讨论，

然后再阅读你的结果，希望能够了解你的结果的含义以及你的论文的重要性，而无需学习如何解读你的实验结果。他们也可能想知道你对你所在领域未来的看法。更熟悉你所在领域的读者通常能够理解你的实验结果，但他们会好奇你是如何解读那些令人困惑、矛盾或复杂的结果的。

在撰写讨论时，要确定哪些人会对每一段内容感兴趣，以及你希望他们从中获得什么。成功的讨论可以同时提供专家想要阅读的具体、细致的信息，以及非专家也能理解的更广泛、更概括的陈述。

目标受众中专家和非专家读者的比例取决于你投稿的期刊。知名度高、面向大众的期刊会有更多的非专家读者，而技术性更强、领域特定的期刊几乎完全由专家读者构成。

技能

讲述你的论文有何特别之处

弱讨论部分以结果总结或引言要点的重复开头。强讨论部分则能通过阐述本文的有趣之处或独特之处，在浩瀚的论文中脱颖而出。一种方法是用一两句话开始讨论，阐述研究结果的主要发现及其对该领域的意义。

将您的结果与现有结果联系起来

在引言中，你可能通过引用该领域先前的研究成果来激发你的研究动机。既然你已经阐述了你的研究结果，你应该说明你的研究结果是否与先前的研究一致，并说明原因。你可能扩展了先前的研究，展示了看似矛盾的结果实际上是和谐的，或者揭示了一个目前尚无解释的矛盾之处。

说明你的研究的局限性如何留下一些大问题

每项研究都是有限的：你做了一些事情，而没有做其他事情，你使用的方法可以解释某些现象，而不能解释其他现象。你的研究的局限性如何留下了更大的问题？你是否只需要做更多同类的工作？你是否已经证明，目前的方法不足以回答这个大问题？

每篇论文都是对更广阔的科学讨论的贡献。希望您认为您的贡献在某种程度上对这场讨论有所助益：它提供了新的信息或工具，能够帮助您或其他研究人员找到重大问题的答案。

为了解释这一贡献，许多讨论会以前瞻性陈述结尾，试图将论文置于该领域研究的预期未来发展中。

带注释的示例：

DISCUSSION

big summary/ importance statement for the science

These results show that the capacity to induce T_{regs} and modify their phenotype is a characteristic of more effector strains than was appreciated previously.

scientific findings as extension of previous work

Our findings concerning the role of human gut bacteria in shaping features of the gut mucosal immune system complement and extend the elegant work by Atarashi *et al.* (*16*). They used a single selective condition (chloroform treatment) to recover a group of 17 strains (all of which were described as members of the class Clostridia) from the human fecal microbiota of a single donor and showed that the consortium was capable of expanding the colonic regulatory T cell compartment in gnotobiotic mice.

findings' scientific implications

The fact that we found this effector activity among gut species belonging to other bacterial phyla suggests that distribution of this functional capacity may be beneficial in ensuring that this tolerogenic cell type is consistently and persistently maintained in different microbial community and host contexts.

how this engineering will facilitate future research

The approach we describe allows systematic follow-up analyses of the extent to which the T_{reg} response is affected by factors such as age at colonization or by different diets that produce abrupt and substantial alterations in microbiota configurations (*45–47*). Despite identifying members of different human gut bacterial phyla that shape the T_{reg} response, our study and that of Atarashi *et al.* revealed that intestinal short-chain fatty acid concentrations increased upon colonization. Given the substantial amount of data supporting a role for short-chain fatty acids in the induction of T_{regs} (*42–44*), this suggests a common pathway by which different microbes converge to modulate this facet of the host immune system. The genetic manipulability of some of the bacterial strains identified here, notably the *Bacteroides*, affords an opportunity to test this and other hypotheses, and advance our knowledge about the molecular underpinnings of microbiota-T_{reg} crosstalk.

forward-looking statements about the field as a whole

As the field of human microbial ecology research moves from observational studies to hypothesis-driven experiments designed to directly test the contributions of the microbiota and its components to health, there is a growing need to develop and transition to a modernized set of Koch's postulates (*48*) where the groups of microbes that modulate host phenotypic responses are identified along with the environmental factors (for example, dietary) necessary for the response to be fully manifest.

a summary/ importance of the engineering aspect of the work

We have developed a platform for systematically identifying microbe-host phenotype interactions in different (human) donor microbiota using gnotobiotic mice that can represent different host genetic features and different environmental conditions of interest. With the 17 strains in our culture collection, there were more than 100,000 possible combinations to search for effector strains. Using the mathematical and experimental strategies described, we only needed 100 combinations to identify multiple effector microbes for three very diverse biological responses (metabolic, adiposity, and T_{regs}).

how it facilitates scientific research

This represents a 1000-fold reduction in the search space compared to what would be required theoretically. By testing these 100 combinations of microbes in an out-of-the-isolator gnotobiotic caging system rather than in traditional flexible film isolators, we overcame what would have been an insurmountable practical barrier to performing these studies for most groups. Our entire study could have been completed with a single flexible film isolator to generate the required germ-free mice. This feature suggests that our overall approach should be accessible to many investigators because animal facilities with small numbers of gnotobiotic isolators already exist in numerous universities.

limitations of the platform show which questions would be unanswered

Although identifying effector strains represents a critical first step in mechanistic analyses of how the gut microbiota affects various facets of host biology, once such strains are identified, much additional work needs to be done. For example, numerous other important components of the intestinal immune system may also be affected by colonization with the strains we identified, including B cell class switching to IgA, macrophage/dendritic cell effector or migratory properties, and $\gamma\delta$ T cell function. Another important goal is to identify the effector molecules produced by the identified effector strains and the host signaling pathways through which these molecules act. Using gnotobiotic mice genetically deficient in various components of the immune system (such as Toll-like receptors or inflammasomes) and effector strains that are genetically manipulated (for example, through whole-genome transposon mutagenesis) represent ways for pursuing this goal. Although additional elements of these mechanistic analyses will be dependent on the biological processes being interrogated, in principle this platform can be applied to any microbiota-associated phenotype.

how it will facilitate medical research

Finally, our approach has therapeutic implications because it represents an enabling system for identifying and characterizing next-generation probiotics or combinations of pre- and probiotics (synbiotics).

Faith *et al.*, *Science Translational Medicine* (2014)
doi:10.1126/scitranslmed.3008051

DISCUSSION

summary of results

In this work, we characterize Cpf1-containing class 2 CRISPR systems, classified as type V, and show that its effector protein, Cpf1, is a single RNA-guided endonuclease. Cpf1 substantially differs from Cas9—to date, the only other experimentally characterized class 2 effector—in terms of structure and function and might provide important advantages for genome-editing applications. Specifically, Cpf1 contains a single identified nuclease domain, in contrast to the two nuclease domains present in Cas9. The results presented here show that, in FnCpf1, inactivation of RuvC-like domain abolishes cleavage of both DNA strands. Conceivably, FnCpf1 forms a homodimer (Figure S2B), with the RuvC-like domains of each of the two subunits cleaving one DNA strand. However, we cannot rule out that FnCpf1 contains a second yet-to-be-identified nuclease domain. Structural characterization of Cpf1-RNA-DNA complexes will allow testing of these hypotheses and elucidation of the cleavage mechanism.

limitation of study

Perhaps the most notable feature of Cpf1 is that it is a single crRNA-guided endonuclease. Unlike Cas9, which requires tracrRNA to process crRNA arrays and both crRNA and tracrRNA to mediate interference (Deltcheva et al., 2011), Cpf1 processes crRNA arrays independent of tracrRNA, and Cpf1-crRNA complexes alone cleave target DNA molecules, without the requirement for any additional RNA species. This feature could simplify the design and delivery of genome-editing tools. For example, the shorter (~42 nt) crRNA employed by Cpf1 has practical advantages over the long (~100 nt) guide RNA in Cas9-based systems because shorter RNA oligos are significantly easier and cheaper to synthesize. In addition, these findings raise more fundamental questions regarding the guide processing mechanism of the type V CRISPR-Cas systems. In the case of type II, processing of the pre-crRNA is catalyzed by the bacterial RNase III, which recognizes the long duplex formed by the tracrRNA and the complementary portion of the direct repeat (Deltcheva et al., 2011). Such long duplexes are not present in the pre-crRNA of type V systems, making it unlikely that RNase III is responsible for processing. Further experiments aimed at elucidating the processing mechanism of type V systems will shed light on the functional diversity of different CRISPR-Cas systems.

engineering implications

scientific implications

Cpf1 generates a staggered cut with a 5′ overhang, in contrast to the blunt ends generated by Cas9 (Garneau et al., 2010; Jinek et al., 2012; Gasiunas et al., 2012). This structure of the cleavage product could be particularly advantageous for facilitating non-homologous end joining (NHEJ)-based gene insertion into the mammalian genome (Maresca et al., 2013). Being able to program the exact sequence of a sticky end would allow researchers to design the DNA insert so that it integrates into the genome in the proper orientation. Specifically, in non-dividing cells, in which genome editing via homology-directed repair (HDR) mechanisms is especially challenging (Chan et al., 2011), Cpf1 could provide an effective way to precisely introduce DNA into the genome via non-HDR mechanisms.

more engineering implications

Another potentially useful feature of Cpf1 that might aid the introduction of new DNA sequences is that Cpf1 cleaves target DNA at the distal end of the protospacer, far away from the seed region. Therefore, Cpf1-induced indels will be located far from the target site, which is thus preserved for subsequent rounds of Cpf1 cleavage. With Cas9, any indel resulting from the dominant NHEJ repair pathway will disrupt the target site, effectively eliminating the possibility of inserting new DNA at that site in that particular cell. In the case of Cpf1, it appears possible that, if the first round of targeting results in an indel, a subsequent round of targeting could yet be repaired via HDR. Future exploration of these and other strategies using Cpf1 and other class 2 effectors is expected to bring solutions for some of the biggest challenges facing genome editing.

The T-rich PAMs of the Cpf1-family also allow for applications in genome editing in organisms with particularly AT-rich genomes, such as *Plasmodium falciparum* (Gardner et al., 2002) or areas of interest with AT enrichment, such as scaffold/matrix attachment regions. To date, all characterized mammalian genome-editing proteins require the presence of at least one G (Hsu et al., 2014; Jiang et al., 2015), so the T- and T/C-dependent PAMs of Cpf1-family proteins expand the targeting range of RNA-guided genome editing nucleases.

The natural diversity of CRISPR systems provides a wealth of opportunities for understanding the origin and evolution of prokaryotic adaptive immunity, as well as for harnessing potentially transformative biotechnological tools. There is little doubt that, beyond the already classified and characterized diversity of the CRISPR-Cas types, there are additional systems with distinctive characteristics that await exploration and could further enhance genome editing and other areas of biotechnology as well as shed further light on the evolution of these defense systems.

summary of implications

forward-looking statement

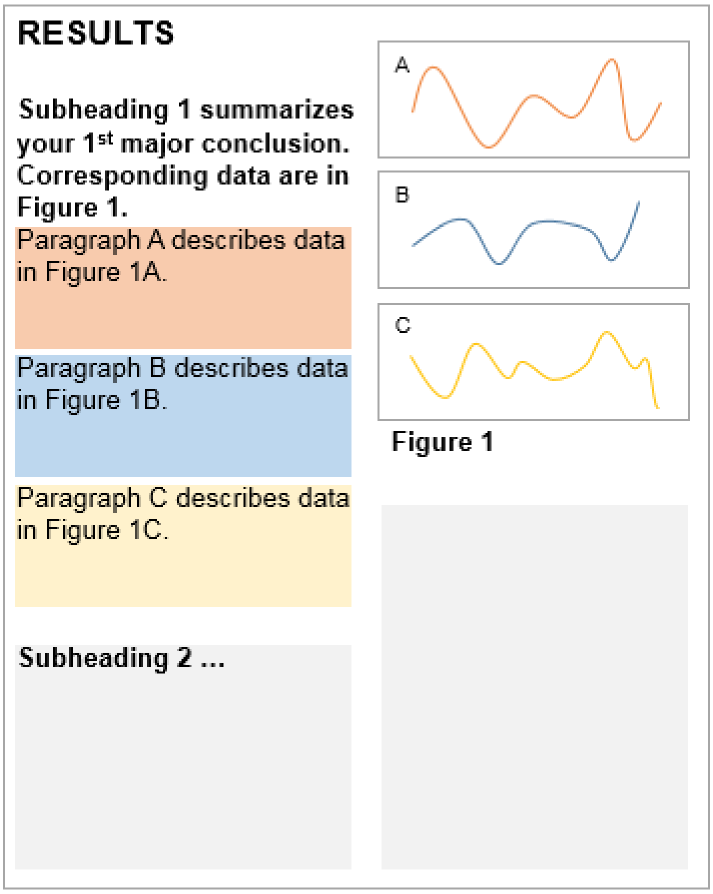
Cell *163*, 759–771, October 22, 2015 ©2015 Elsevier Inc.

结论

成功标准

1. 数据和从中得出的结论描述清晰，没有任何推测。
2. 数据以叙述流的形式呈现，没有逻辑跳跃。
3. 这个故事逐渐形成了一条值得深思的信息。

结构图



另外，比较方法和结果的真实注释示例：注意两个部分之间的副标题的对应性。

注意：

"Results" 与 "Discussion" 是否分开写？

不同 SCI 期刊在 Author Guidelines 中会明确说明格式要求。

如 Nature、Science、Cell 等高水平综合类期刊倾向于将 Results 与 Discussion 合并，注重叙述的流畅性与整体逻辑；

工程类、材料类、医学类大多数期刊（如 ACS 系列、Elsevier 系列、Springer 系列）多

为分开结构，便于审稿人快速理解实验过程与逻辑推导。

部分	主要内容	写作风格	内容性质
Results	只陈述观察结果、数据图表、趋势	客观、中性、不解释	事实
Discussion	解释结果、推导机制、引用文献、比较他人研究	推理、逻辑、引证	主观推断+科学解释
Conclusions	总结关键发现、科学贡献、研究意义；可能展望未来	概括、精炼、有总结性	提炼升华

期刊要求分开写：Introduction → Methods → Results → Discussion → Conclusions

期刊允许合并写：Introduction → Methods → Results and Discussion → Conclusions

目的

结果部分的目标是：

1. 以叙述的形式尽可能客观地描述和解释你用你的方法获得的数据。
2. 根据这些数据传达一个值得注意的信息。

虽然图表让读者有机会直接检查数据并得出自己的结论，但结果部分通过解释实验逻辑、强调重要数据特征和陈述结论，为解释数据的过程提供了更多支持。

推测和扩展解释属于讨论，而不是结果。

分析你的受众

论文的结果部分适用于那些不敢仅根据数字分析数据和结论的读者，或者除了视觉信息外还喜欢口头信息的读者。

某些类型的读者可能会直接从您的图表中得出结论，只是偶尔参考您的结果：他们要么时间紧迫，要么是您子领域的专家，因此习惯于解读与您相同类型的数据（例如，不仅是生物工程师，还有系统生物学家或组织工程师）。然而，许多其他读者依靠结果来帮助他们更深入

地了解您的研究成果和研究成果。

尤其对于非专业人士来说，您在“结果”部分提供的解释将使其受益匪浅。因此，不要假设读者知道特定方法的目的，也不要假设特定观察的含义是不言而喻的：请尽可能清晰地阐述您的动机和解释。此外，如果可以使用更简单的术语，请避免使用特定领域的术语。

您可以请不同实验室的朋友阅读您的结果并帮助您识别无法解释的术语或需要更清楚地说明理由的地方。

技能

创建一个逻辑叙述，并组织成小节

为你的成果找到叙述顺序的一个好方法是先整理你的图表。在撰写成果之前，你应该已经确定了论文中要包含的图表。每个图表都应支持一个特定的结论，并提供读者评估该结论所需的数据（参见图表）。重新排列你的图表，直到你找到一个能够形成最合乎逻辑的一系列结论的顺序，从而得出最终的成果。用这一系列图表/结论作为你的成果的提纲。

每个主要结论（可能对应一张或多张图表）都可以作为“结果”部分子章节的标题。这种模块化结构有助于读者快速将图表与“结果”以及“结果”进行匹配，从而轻松浏览您的论文。（另请参阅“方法”部分中的“使用副标题组织内容” ）。

每个结果段落都有理由、数据和过渡

单个“结果”段落通常对应单个实验（或一组密切相关的实验），以及单个图表或较大图表中的子图。（参见上文的结构图。）

1. 每个结果段落都以一个主题句开头，解释进行实验的理由。例如，你可以使用如下结构：“为了确定 X，进行了 Y 项研究，并显示出 Z 主要结果。”
2. 在主题句之后，按逻辑顺序描述您的数据和从中得出的结论：例如，先赞成再反对，从最重要到最不重要，实验组与对照组。
3. 最后用一个过渡句来总结研究结果，并在必要时解释为什么进行下一个实验或假设。

只有在需要解释实验之间的转变时，才可以在结果中包含推测：

1. “通过观察数据 A，我们推测机制 B 可能导致现象 Z。因此，下一个实验通过以下方式测试机制 B 的活性……”

显示最少的必要数据

描述所有必要的数据，以便读者评估你的结论，仅此而已。强迫读者分析不必要的细节会分散他们对你主要信息的注意力。不要包含与给定结论无关的数据。确定哪些数据是相关的可能很棘手，通常需要一些判断。这也取决于期刊文章的长度限制——你的文章越短，你可能需要将更多内容移到补充信息中，或者省略掉。

以下是一些帮助您决定包含和排除哪些数据的一般准则：

包括	填写补充信息，或排除
最有力地证明您结论的实验或数据集。 • 请注意，您选择的数据集的某些部分可能与您的主要结论相矛盾，或者支持您的某个主张但不支持其他主张。 • 在描述任何此类矛盾时要清晰、诚实，特别是如果它们可能反映出读者在评估你的主要主张时应该知道的局限性（例如，方法论上的缺陷、对生物系统的理解不完整）。	• 实验或数据集也支持你的结论，但不是最有力的证据 例如，该方法的验证程度较低，数据统计意义不大，或者数据解释起来不太直观 • 为验证方法而进行的实验 • 为排除其他假设而进行的实验

描述单个结果所花费的时间应与该结果对论文主要结论的重要性成正比。描述复杂或令人困惑的结果时，很容易写得更多，但你也不想让读者的脑子里塞满各种细节，以免分散他们对主要结论的注意力。在写作过程中，要不断提醒自己哪些结论最为重要，并为支持这些结论的细节留出最多的篇幅和细节。

使用适当的风格

结果应以过去时态书写。

尽量保持客观。除了避免推测之外，还要避免使用“有趣的是，我们发现……”之类的说法，除非这种“有趣”能够得到具体证明——例如，结果与某个主要假设或该领域以往的研究结果相矛盾。

Cell

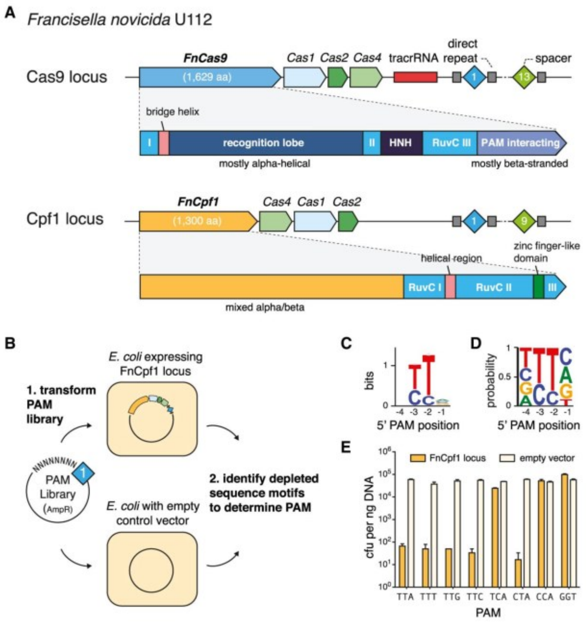


Figure 1. The *Francisella novicida U112* Cpf1 CRISPR Locus Provides Immunity against Transformation of Plasmids Containing Protospacers Flanked by a 5′-TTN PAM

(A) Organization of two CRISPR loci found in *Francisella novicida U112* (NC_008601). The domain architectures of FnCas9 and FnCpf1 are compared.

(B) Schematic illustrating the plasmid depletion assay for discovering the PAM position and identity. Competent *E. coli* harboring either the heterologous FnCpf1 locus plasmid (pFnCpf1) or the empty vector control were transformed with a library of plasmids containing the matching protospacer flanked by randomized 5′ or 3′ PAM sequences and selected with antibiotic to deplete plasmids carrying successfully targeted PAM. Plasmids from surviving colonies were extracted and sequenced to determine depleted PAM sequences.

(C and D) Sequence logo for the FnCpf1 PAM as determined by the plasmid depletion assay. Letter height at each position is measured by information content (C) or frequency (D); error bars show 95% Bayesian confidence interval.

(E) *E. coli* harboring pFnCpf1 provides robust interference against plasmids carrying 5′-TTN PAMs (n = 3; error bars represent mean ± SEM).

See also Figure S1.

been recently identified in several bacterial genomes (http://www.jcvi.org/cgi-bin/tigrfams/HmmReportPage.cgi?acc=TIGR04330) (Schunder et al., 2013; Vestergaard et al., 2014; Makarova et al., 2015). The putative type V CRISPR-Cas systems contain a large, ~1,300 amino acid protein called Cpf1 (CRISPR from *Prevotella* and *Francisella* 1). It remains unknown, however, whether Cpf1-containing CRISPR loci indeed represent functional CRISPR systems. Given the broad applications of Cas9 as a genome-engineering tool (Hsu et al., 2014; Jiang and Marraffini, 2015), we sought to explore the function of Cpf1-based putative CRISPR systems.

Here, we show that Cpf1-containing CRISPR-Cas loci of *Francisella novicida U112* encode functional defense systems capable of mediating plasmid interference in bacterial cells guided by the CRISPR spacers. Unlike Cas9 systems, Cpf1-containing CRISPR systems have three features. First, Cpf1-associated CRISPR arrays are processed into mature crRNAs without the requirement of an additional *trans*-activating crRNA (tracrRNA) (Deltcheva et al., 2011; Chylinski et al., 2013). Second, Cpf1-crRNA complexes efficiently cleave target DNA proceeded by a short T-rich protospacer-adjacent motif (PAM), in contrast to the G-rich PAM following the target DNA for Cas9 systems. Third, Cpf1 introduces a staggered DNA double-stranded break with a 4 or 5-nt 5′ overhang.

To explore the suitability of Cpf1 for genome-editing applications, we characterized the RNA-guided DNA-targeting requirements for 16 Cpf1-family proteins from diverse bacteria, and we identified two Cpf1 enzymes from *Acidaminococcus sp. BV3L6* and *Lachnospiraceae bacterium ND2006* that are capable of mediating robust genome editing in human cells. Collectively, these results establish Cpf1 as a class 2 CRISPR-Cas system that includes an effective single RNA-guided endonuclease with distinct properties that has the potential to substantially advance our ability to manipulate eukaryotic genomes.

RESULTS

Cpf1-Containing CRISPR Loci Are Active Bacterial Immune Systems

Subheading states conclusion

Cpf1 was first annotated as a CRISPR-associated gene in TIGRFAM (http://www.jcvi.org/cgi-bin/tigrfams/HmmReportPage.cgi?acc=TIGR04330) and has been hypothesized to be the effector of a CRISPR locus that is distinct from the Cas9-containing type II CRISPR-Cas loci that are also present in the genomes of some of the same bacteria, such as multiple strains of *Francisella* and *Prevotella* (Schunder et al., 2013; Vestergaard et al., 2014; Makarova et al., 2015) (Figure 1A). The Cpf1 protein contains a predicted RuvC-like endonuclease domain that is distantly related to the respective nuclease domain of Cas9. However, Cpf1 differs from Cas9 in that it lacks a second, HNH endonuclease domain, which is inserted within the

Section content corresponds to Fig. 1 (above)

If you think your data really prove something definitively, and it's important for your reader to know it, you can use subjective terms like "clearly."

RuvC-like domain of Cas9. Furthermore, the N-terminal portion of Cpf1 is predicted to adopt a mixed α/β structure and appears to be unrelated to the N-terminal, α-helical recognition lobe of Cas9 (Figure 1A). It has been shown that the nuclease moieties of Cas9 and Cpf1 are homologous to distinct groups of transposon-encoded TnpB proteins, the first one containing both RuvC and HNH nuclease domains and the second one containing the RuvC-like domain only (Makarova and Koonin, 2015). Apart from these distinctions between the effector proteins, the Cpf1-carrying loci encode Cas1, Cas2, and Cas4 proteins that are more closely related to orthologs from types I and III than to those from type II CRISPR systems (Makarova et al., 2015). Taken together, these differences from type II have prompted the classification of Cpf1-encoding CRISPR-Cas loci as the putative type V within class 2 (Makarova et al., 2015). The features of the putative type V loci, especially the domain architecture of Cpf1, suggest not only that type II and type V systems independently evolved through the association of different adaptation modules (*cas1*, *cas2*, and *cas4* genes) with different TnpB genes, but also that type V systems are functionally unique. The notion that Cpf1-carrying loci are bona fide CRISPR systems is further buttressed by the search of microbial genome sequences for similarity to the type V spacers that produced several significant hits to prophage genes—in particular, those from *Francisella* (Schunder et al., 2013). Given these observations and the prevalence of Cpf1-family proteins in diverse bacterial species, we sought to test the hypothesis that Cpf1-encoding CRISPR-Cas loci are biologically active and can mediate targeted DNA interference, one of the primary functions of CRISPR systems.

Experimental rationale + methods description without too much detail

To simplify experimentation, we cloned the *Francisella novicida U112* Cpf1 (FnCpf1) locus (Figure 1A) into low-copy plasmids (pFnCpf1) to allow heterologous reconstitution in *Escherichia coli*. Typically, in currently characterized CRISPR-Cas systems, there are two requirements for DNA interference: (1) the target sequence has to match one of the spacers present in the respective CRISPR array, and (2) the target sequence complementary to the spacer (hereinafter protospacer) has to be flanked by the appropriate protospacer adjacent motif (PAM). Given the completely uncharacterized functionality of the FnCpf1 CRISPR locus, we adapted a previously described plasmid depletion assay (Jiang et al., 2013) to ascertain the activity of Cpf1 and identify the requirement for a PAM sequence and its respective location relative to the protospacer (5′ or 3′) (Figure 1B). We constructed two libraries of plasmids carrying a protospacer matching the first spacer in the FnCpf1 CRISPR array with the 5′ or 3′ 7 bp sequences randomized. Each plasmid library was transformed into *E. coli* that heterologously expressed the FnCpf1 locus or into a control *E. coli* strain carrying the empty vector. Using this assay, we determined the PAM sequence and location by identifying nucleotide motifs that are preferentially depleted in cells heterologously expressing the FnCpf1 locus. We found that the PAM for FnCpf1 is located upstream of the 5′ end of the displaced strand of the protospacer and has the sequence 5′-TTN (Figures 1C, 1D and S1). The 5′ location of the PAM is also observed in type I CRISPR systems, but not in type II systems, where Cas9 employs PAM sequences that are located on the 3′ end of the protospacer (Mojica et al., 2009; Garneau et al., 2010). Beyond the identification of the PAM, the results of the depletion assay clearly indicate that heterologously expressed Cpf1 loci are capable of efficient interference with plasmid DNA.

Quick description of how assay was interpreted

Statement of findings

In this paper, comparison with previously discovered systems is important for establishing what the findings mean.

Conclusion

Transition + rationale + methods

Findings

Conclusion

To further characterize the PAM requirements, we analyzed plasmid interference activity by transforming *cpf1*-locus-expressing cells with plasmids carrying protospacer 1 flanked by 5′-TTN PAMs. We found that all 5′-TTN PAMs were efficiently targeted (Figure 1E). In addition, 5′-CTA, but not 5′-TCA, was also efficiently targeted (Figure 1E), suggesting that the middle T is more critical for PAM recognition than the first T and that, in agreement with the sequence motifs depleted in the PAM discovery assay (Figure S1D), the PAM might be more relaxed than 5′-TTN.

The Cpf1-Associated CRISPR Array Is Processed Independent of TracrRNA

Section content corresponds to Fig. 2 (and so on for following sections)

After showing that *cpf1*-based CRISPR loci are able to mediate robust DNA interference, we performed small RNA sequencing to determine the exact identity of the crRNA produced by these loci. By sequencing small RNAs extracted from a *Francisella novicida U112* culture, we found that the CRISPR array is processed into short mature crRNAs of 42–44 nt in length. Each mature crRNA begins with 19 nt of the direct repeat followed by 23–25 nt of the spacer sequence (Figure 2A). This crRNA arrangement contrasts with that of type II CRISPR-Cas systems in which the mature crRNA starts with 20–24 nt of spacer sequence followed by ~22 nt of direct repeat (Deltcheva et al., 2011; Chylinski et al., 2013). Unexpectedly, apart from the crRNAs, we did not observe any robustly expressed small transcripts near the *Francisella cpf1* locus that might correspond to tracrRNAs, which are associated with Cas9-based systems.

To confirm that no additional RNAs are required for crRNA maturation and DNA interference, we constructed an expression plasmid using synthetic promoters to drive the expression of *Francisella cpf1* (FnCpf1) and the CRISPR array (pFnCpf1_min). Small RNAseq of *E. coli* expressing this plasmid still showed robust processing of the CRISPR array into mature crRNA (Figure 2B), indicating that FnCpf1 and its CRISPR array are the only elements required from the FnCpf1 locus to achieve crRNA processing. Furthermore, *E. coli* expressing pFnCpf1_min as well as pFnCpf1_ΔCas, a plasmid with all of the *cas* genes removed but retaining native promoters driving the expression of FnCpf1 and the CRISPR array, also exhibited robust DNA interference, demonstrating that FnCpf1 and crRNA are sufficient for mediating DNA targeting (Figure 2C). By contrast, Cas9 requires both crRNA and tracrRNA to mediate targeted DNA interference (Deltcheva et al., 2011; Zhang et al., 2013).

Cpf1 Is a Single crRNA-Guided Endonuclease

The finding that FnCpf1 can mediate DNA interference with crRNA alone is highly surprising given that Cas9 recognizes crRNA through the duplex structure between crRNA and tracrRNA (Jinek et al., 2012; Nishimasu et al., 2014), as well as the 3′ secondary structure of the tracrRNA (Hsu et al., 2013; Nishimasu et al., 2014). To ensure that crRNA is indeed sufficient for forming an active complex with FnCpf1 and mediating RNA-guided DNA cleavage, we investigated whether FnCpf1 supplied only with crRNA can cleave target DNA in vitro. We purified

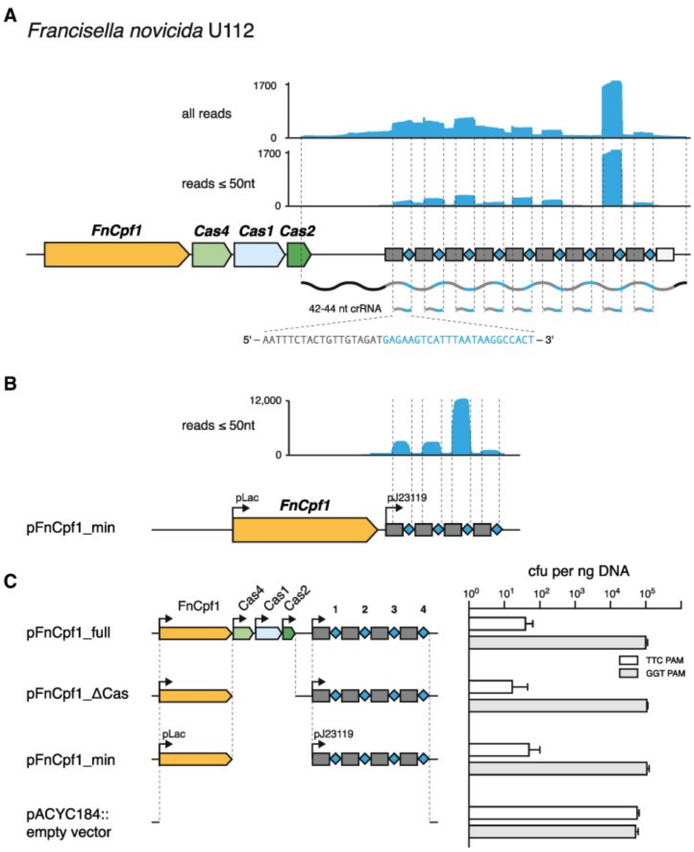


Figure 2. Heterologous Expression of FnCpf1 and CRISPR Array in *E. coli* Is Sufficient to Mediate Plasmid DNA Interference and crRNA Maturation

(A) Small RNA-seq of *Francisella novicida U112* reveals transcription and processing of the FnCpf1 CRISPR array. The mature crRNA begins with a 19-nt partial direct repeat followed by 23–25 nt of spacer sequence.

(B) Small RNA-seq of *E. coli* transformed with a plasmid-carrying synthetic promoter-driven FnCpf1 and CRISPR array shows crRNA processing independent of Cas genes and other sequence elements in the FnCpf1 locus.

(C) *E. coli* harboring different truncations of the FnCpf1 CRISPR locus shows that only FnCpf1 and the CRISPR array are required for plasmid DNA interference (n = 3; error bars show mean ± SEM).

FnCpf1 (Figure S2) and assayed its ability to cleave the same protospacer-1-containing plasmid used in the bacterial DNA interference experiments (Figure 3A). We found that FnCpf1 along with an in-vitro-transcribed mature crRNA-targeting protospacer 1 was able to efficiently cleave the target plasmid in a Mg^{2+}- and crRNA-dependent manner (Figure 3B). Moreover, FnCpf1 was able to cleave both supercoiled and linear target DNA (Figure 3C). These results clearly demonstrate the sufficiency of FnCpf1 and crRNA for RNA-guided DNA cleavage.

We also mapped the cleavage site of FnCpf1 using Sanger sequencing of the cleaved DNA ends. We found that FnCpf1-mediated cleavage results in a 5-nt 5′ overhang (Figures 3A, 3D, and S3A–S3D), which is different from the blunt cleavage product generated by Cas9 (Garneau et al., 2010; Jinek et al., 2012; Gasiunas et al., 2012). The staggered cleavage site of FnCpf1 is distant from the PAM: cleavage occurs after the 18th base on the non-targeted (+) strand and after the 23rd base on the targeted (–) strand (Figures 3A, 3D, and S3A–S3D). Using double-stranded oligo substrates with different PAM sequences, we also found that FnCpf1 requires the 5′-TTN PAM to be in a duplex form in order to cleave the target DNA (Figure 3E).

The RuvC-like Domain of Cpf1 Mediates RNA-Guided DNA Cleavage

The RuvC-like domain of Cpf1 retains all of the catalytic residues of this family of endonucleases (Figures 4A and S4) and is thus predicted to be an active nuclease. Therefore, we generated three mutants—FnCpf1(D917A), FnCpf1(E1006A), and FnCpf1(D1225A) (Figure 4A)—to test whether the conserved catalytic residues are essential for the nuclease activity of FnCpf1. We found that the D917A and E1006A mutations completely inactivated the DNA cleavage activity of FnCpf1, and D1255A significantly reduced nucleolytic activity (Figure 4B). These results are in contrast to the mutagenesis results for *Streptococcus pyogenes* Cas9 (SpCas9), where mutation of the RuvC (D10A) and HNH (N863A) nuclease domains converts SpCas9 into a DNA nickase (i.e., inactivation of each of the two nuclease domains abolished the cleavage of one of the DNA strands) (Jinek et al., 2012; Gasiunas et al., 2012) (Figure 4B). These findings suggest that the RuvC-like domain of FnCpf1 cleaves both strands of the target DNA, perhaps in a dimeric configuration. Interestingly, size-exclusion gel filtration of FnCpf1 shows that the protein is eluted at a size of ~300 kD, twice the molecular weight of a FnCpf1 monomer (Figure S2B).

Sequence and Structural Requirements for the Cpf1 crRNA

Compared with the guide RNA for Cas9, which has elaborate RNA secondary structure features that interact with Cas9 (Nishimasu et al., 2014), the guide RNA for FnCpf1 is notably simpler and only consists of a single stem loop in the direct repeat

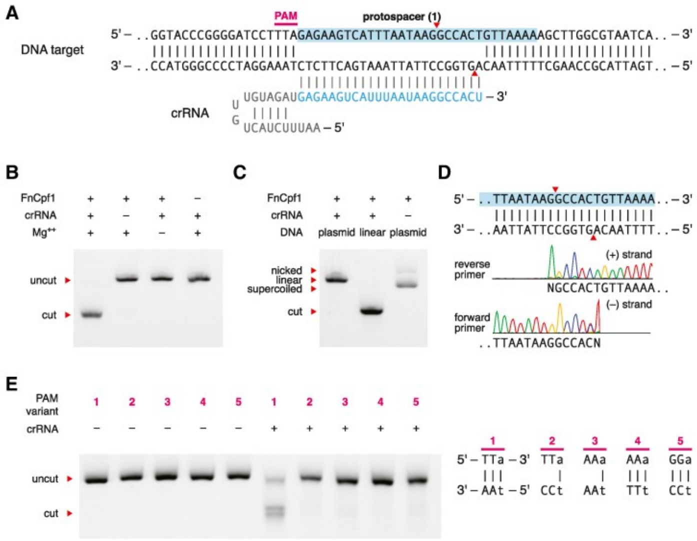


Figure 3. FnCpf1 Is Guided by crRNA to Cleave DNA In Vitro

(A) Schematic of the FnCpf1 crRNA-DNA-targeting complex. Cleavage sites are indicated by red arrows.

(B) FnCpf1 and crRNA alone mediated RNA-guided cleavage of target DNA in a crRNA- and Mg^{2+}-dependent manner.

(C) FnCpf1 cleaves both linear and supercoiled DNA.

(D) Sanger-sequencing traces from FnCpf1-digested target show staggered overhangs. The non-templated addition of an additional adenine, denoted as N, is an artifact of the polymerase used in sequencing (Clark, 1988). Reverse primer read represented as reverse complement to aid visualization. See also Figure S3.

(E) Dependency of cleavage on base-pairing at the 5′ PAM. FnCpf1 can only recognize the PAM in correctly Watson-Crick-paired DNA.

See also Figures S2 and S3.

sequence (Figure 3A). We explored the sequence and structural requirements of crRNA for mediating DNA cleavage with FnCpf1.

We first examined the length requirement for the guide sequence and found that FnCpf1 requires at least 16 nt of guide sequence to achieve detectable DNA cleavage and a minimum of 18 nt of guide sequence to achieve efficient DNA cleavage in vitro (Figure 5A). These requirements are similar to those demonstrated for SpCas9, in which a minimum of 16–17 nt of spacer sequence is required for DNA cleavage (Cencic et al., 2014; Fu et al., 2014). We also found that the seed region of the FnCpf1 guide RNA is approximately within the first 5 nt on the 5′ end of the spacer sequence (Figures 5B and S3E).

Next, we studied the effect of direct repeat mutations on the RNA-guided DNA cleavage activity. The direct repeat portion of mature crRNA is 19 nt long (Figure 2A). Truncation of the direct repeat revealed that at least 16, but optimally more than 17 nt, of the direct repeat is required for cleavage. Mutations in the stem loop that preserved the RNA duplex did not affect the cleavage activity, whereas mutations that disrupted the stem loop duplex structure completely abolished cleavage (Figure 5D). Finally, base substitutions in the loop region did not affect nuclease activity, whereas the uracil base immediately proceeding the spacer sequence could not be substituted (Figure 5E). Collectively, these results suggest that FnCpf1 recognizes the crRNA through a combination of sequence-specific and structural features of the stem loop.

Cpf1-Family Proteins from Diverse Bacteria Share Common crRNA Structures and PAMs

Based on our previous experience in harnessing Cas9 for genome editing in mammalian cells, only a small fraction of bacterial nucleases can function efficiently when heterologously expressed in mammalian cells (Cong et al., 2013; Ran et al., 2015). Therefore, in order to assess the feasibility of harnessing Cpf1 as a genome-editing tool, we exploited the diversity of Cpf1-family proteins available in the public sequences databases. A BLAST search of the WGS database at the NCBI revealed 46 non-redundant Cpf1-family proteins (Figure S5A), from which we chose 16 candidates that, based on our phylogenetic reconstruction (Figure S5A), represented the entire Cpf1 diversity (Figures 6A and S5). These Cpf1-family proteins span a range of lengths between ∼1,200 and ∼1,500 amino acids.

The direct repeat sequences for each of these Cpf1-family proteins show strong conservation in the 19 nt at the 3′ of the direct repeat, the portion of the repeat that is included in the processed crRNA (Figure 6B). The 5′ sequence of the direct repeat is much more diverse. Of the 16 Cpf1-family proteins chosen for analysis, three (2, *Lachnospiraceae bacterium MC2017*, Lb3Cpf1; 3, *Butyrivibrio proteoclasticus*, BpCpf1; and 6, *Smithella sp. SC_K08D17*, SsCpf1) were associated with direct repeat sequences that are notably divergent from the FnCpf1 direct repeat (Figure 6B). However, even these direct repeat sequences preserved stem-loop structures that were identical or nearly identical to the FnCpf1 direct repeat (Figure 6C).

Given the strong structural conservation of the direct repeats that are associated with many of the Cpf1-family proteins, we first tested whether the orthologous direct repeat sequences are able to support FnCpf1 nuclease activity in vitro. As expected, the direct repeats that contained conserved stem sequences were able to function interchangeably with FnCpf1. By contrast, the direct repeats from candidates 2 (Lb3Cpf1) and 6 (SsCpf1) were unable to support FnCpf1 cleavage activity (Figure 6D). The direct repeat from candidate 3 (BpCpf1) supported only a low level of FnCpf1 nuclease activity (Figure 6D), possibly due to the conservation of the 3′-most U.

Next, we applied the in vitro PAM identification assay (Figure S6A) to determine the PAM sequence for each Cpf1-family protein. We were able to identify the PAM sequence for seven

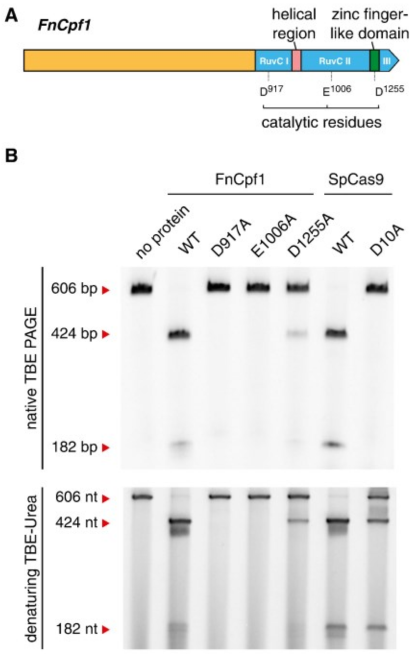


Figure 4. Catalytic Residues in the C-Terminal RuvC Domain of FnCpf1 Are Required for DNA Cleavage

(A) Domain structure of FnCpf1 with RuvC catalytic residues highlighted. The catalytic residues were identified based on sequence homology to *Thermus thermophilus* RuvC (PDB: 4EP5).

(B) Native TBE PAGE gel showing that mutation of the RuvC catalytic residues of FnCpf1 (D917A and E1006A) and mutation of the RuvC (D10A) catalytic residue of SpCas9 prevents double-stranded DNA cleavage. Denaturing TBE-Urea PAGE gel showing that mutation of the RuvC catalytic residues of FnCpf1 (D917A and E1006A) prevents DNA-nicking activity, whereas mutation of the RuvC (D10A) catalytic residue of SpCas9 results in nicking of the target site. See also Figure S4.

new Cpf1-family proteins (Figures 6E, S6B, and S6C), and the screen confirmed the PAM for FnCpf1 as 5′-TTN. The remaining eight tested Cpf1 proteins did not show efficient cleavage during in vitro reconstitution. The PAM sequences for the Cpf1-family proteins were predominantly T rich, only varying in the number of Ts constituting each PAM (Figures 6E, S6B, and S6C).

Cpf1 Can Be Harnessed to Facilitate Genome Editing in Human Cells

We tested each Cpf1-family protein for which we were able to identify a PAM for nuclease activity in mammalian cells. We codon optimized each of these genes and attached a C-terminal nuclear localization signal (NLS) for optimal expression and nuclear targeting in human cells (Figure 7A). To test the activity of each Cpf1-family protein, we selected a guide RNA target site within the *DNMT1* gene (Figure 7B). We first found that each of the Cpf1-family proteins along with its respective crRNA designed to target *DNMT1* was able to cleave a PCR amplicon of the *DNMT1* genomic region in vitro (Figure 7C). However, when tested in human embryonic kidney 293FT (HEK293FT) cells, only two out of the eight Cpf1-family proteins (7, AsCpf1 and 13, LbCpf1) exhibited detectable levels of nuclease-induced indels (Figures 7C and 7D). This result is consistent with previous experiments with Cas9 in which only a small number of Cas9 orthologs were successfully harnessed for genome editing in mammalian cells (Ran et al., 2015).

We further tested each Cpf1-family protein with additional genomic targets and found that AsCpf1 and LbCpf1 consistently mediated robust genome editing in HEK293FT cells, whereas the remaining Cpf1 proteins showed either no detectable activity or only sporadic activity (Figures 7E and S7) despite robust expression (Figure S6D). The only Cpf1 candidate that expressed poorly was PdCpf1 (Figure S6D). When compared to Cas9, AsCpf1 and LbCpf1 mediated comparable levels of indel formation (Figure 7E). Additionally, we used in vitro cleavage followed by Sanger sequencing of the cleaved DNA ends and found that 7, AsCpf1 and 13, LbCpf1 also generated staggered cleavage sites (Figures S6E and S6F, respectively).

DISCUSSION

In this work, we characterize Cpf1-containing class 2 CRISPR systems, classified as type V, and show that its effector protein, Cpf1, is a single RNA-guided endonuclease. Cpf1 substantially differs from Cas9—to date, the only other experimentally characterized class 2 effector—in terms of structure and function and might provide important advantages for genome-editing applications. Specifically, Cpf1 contains a single identified nuclease domain, in contrast to the two nuclease domains present in Cas9. The results presented here show that, in FnCpf1, inactivation of RuvC-like domain abolishes cleavage of both DNA strands. Conceivably, FnCpf1 forms a homodimer (Figure S2B), with the RuvC-like domains of each of the two subunits cleaving one DNA strand. However, we cannot rule out that FnCpf1 contains a second yet-to-be-identified nuclease domain. Structural characterization of Cpf1-RNA-DNA complexes will allow testing of these hypotheses and elucidation of the cleavage mechanism.

Perhaps the most notable feature of Cpf1 is that it is a single crRNA-guided endonuclease. Unlike Cas9, which requires tracrRNA to process crRNA arrays and both crRNA and tracrRNA to mediate interference (Deltcheva et al., 2011), Cpf1 processes crRNA arrays independent of tracrRNA, and Cpf1-crRNA complexes alone cleave target DNA molecules, without the requirement for any additional RNA species. This feature could simplify the design and delivery of genome-editing tools. For example, the shorter (~42 nt) crRNA employed by Cpf1 has practical advantages over the long (~100 nt) guide RNA in Cas9-based systems because shorter RNA oligos are significantly easier and cheaper to synthesize. In addition, these findings raise more fundamental questions regarding the guide processing mechanism of the type V CRISPR-Cas systems. In the case of type II, processing of the pre-crRNA is catalyzed by the bacterial RNase III, which recognizes the long duplex formed by the tracrRNA and the complementary portion of the direct repeat (Deltcheva et al., 2011). Such long duplexes

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应将已提交的稿件或其任何部分上传至生成式 AI 工具或 AI 辅助技术（例如 ChatGPT），因为这可能侵犯作者的保密性和所有权，并且如果论文包含个人身份信息，则可能侵犯数据隐私权。

此保密要求也适用于同行评审报告，因为它可能包含有关稿件和/或作者的机密信息。因此，审稿人不应将其同行评审报告上传到 AI 工具中，即使只是为了改进语言和可读性。

审稿人不应使用生成式人工智能或人工智能辅助技术来协助论文的科学评审。审稿人应对评审报告的内容负责。

技能

以标准方式格式化您的文档

同行评审反馈如果格式清晰、逻辑性强，编辑和作者都更容易消化和理解。最常见的格式是：(1) 摘要，(2) 结论，(3) 主要关注点，以及 (4) 次要关注点（另见上文结构图）。

通常还会有一份选择题表格，用于根据多项标准对论文进行“评分”。编辑可能会根据这份数字评分指南来衡量稿件与其他投稿的优劣；您可以将其视为一份评审中需要涵盖的主题清单。

概括

摘要是综述其余部分的基础。你需要证明你已经阅读并理解了稿件，这有助于作者理解其他读者对稿件主要观点的理解。这也是一个展示你专业知识和批判性思维的机会，这会给编辑留下积极的印象，而编辑往往是你所在领域的重要人物。

使用以下指南会很有帮助：

- 首先用一句话描述论文的主要观点，然后用几句话总结得出论文逻辑结论的具体重要发现。
- 然后，强调论文中显示的重要发现的意义。
- 最后总结审稿人对手稿的优点和缺点的总体看法。

决定

你的决定必须清晰地表述，以便于理解你的其他评论（参见“成功标准”）。你可以将其作为总结段落的结束语，或将其作为总结后的单独句子。一般来说，你可以尝试在以下框架内

进行分类：

- 接受，无需修改
- 接受并进行小修改
- 接受并进行重大修改
- 拒绝

有些期刊会有特定的规则或不同的措辞，因此请确保您了解您的选择。

大多数评审意见都包含向编辑提供保密意见的选项，以便编辑更详细地了解评审结果。在极端情况下，编辑还可以通过此方式提出关于抄袭、数据篡改或其他伦理问题的担忧。

您还可以在决策区域说明您认为自己拥有专业知识来评论复杂手稿的哪些方面。

主要关注点（若相关）

根据您评审的期刊，您的主要关注点可能需要考虑重要性、新颖性、行业相关性或其他特定领域的标准。如果主要关注点严重，通常会导致“拒绝”或“接受并进行重大修改”的决定。

主要关注点包括……

- 论文中提出的论点存在内部不一致的问题，
- 或者提出与该领域重要理解相悖的论点，但没有必要的数据支持。
- 缺乏对证明论文中提出的观点至关重要的关键实验或计算数据。
- Examples: a study that reports the identity of an unexpected peak in a GC-MS spectrum without accounting for common interferences, or claims pertaining to human health when all the data presented is in a model organism or in vitro.

提交包含主要关注点的评论时，最重要的一点是能够引用解决方案。例如……

- 如果您认为某人的论点违反了热力学定律，那么他们需要向您展示什么数据才能让您相信这一点？
- 您需要看到哪些类型的新统计分析才能相信本研究中提出的临床试验的主张？
- 是否需要额外的控制实验来证明这种催化剂确实按照建议的途径促进反应？

小问题（可选）

次要顾虑主要是指那些能够提升信息清晰度，但不会影响论点逻辑的问题。最常见的是……

- 稿件中的语法错误
- 印刷错误
- 缺少参考文献
- 背景或方法信息不足（例如，引言部分仅有五处参考文献）
- 细节不足或可能无关
- 解释不清楚或措辞不当（例如，讨论部分的某个段落似乎与论文的其他部分相矛盾）
- 提高任何图形可读性的可能选项（例如，图形上的错误标签）

虽然在有许多重大问题的审稿中并不总会提到次要问题，但对于被接收或建议做小幅修改后接收的稿件，审稿人几乎总是会列出一些次要问题。

提供合理且在范围内的修订

想想你建议的实验是否可行，以解决你的顾虑。你建议的实验是否需要三年时间才能为另一篇完整的论文奠定基础？如果你建议进行大量的动物实验或测序，那么这些实验的成本会不会高得令人望而却步？如果论文无需进行下一步实验就能立项，那么请认真思考这些额外实验的真正价值。科学出版的主要问题之一是完成论文所需的时间。不要不必要地给其他作者添麻烦！

作为增加实验的替代方案，作者是否需要调整他们的结论，使其更符合现有证据的范围，而不是让证据去迎合结论？如果作者这么做了，这篇论文仍然适合发表在你所审阅的期刊上吗？

以观众能够理解的方式组织你的评论

格式选择：

- 使用换行符（或编号）清楚地分隔每个关注点，并按照它们在手稿中出现的顺序进行组织。
- 直接引用文本，并用粗体或斜体表示相关短语来阐明你的观点
- 包括页码和行/段落编号以便于参考。

风格/简洁性：

- 只需用几句话或更少的句子陈述问题和解决问题的建议，使您的评论尽可能简短。

提供建设性和专业的反馈

保持公正和专业。

虽然在评审过程中，作者的身份有时会保持匿名（这在化学和生物研究中很少见），但研究团体通常规模较小，您可能会根据所用的方法或写作风格“猜测”作者是谁。无论如何，在评审过程中保持公正和专业至关重要。不要根据您对作者身份或他们的研究成果可能对您自身研究产生的影响等看法，对论文做出任何假设。如果您认为这可能对您造成影响，您必须告知编辑，存在利益冲突，您不应评审此稿件。

要有礼貌，要圆滑。

接受批评性反馈，即使是建设性的，对作者来说也可能很困难，甚至可能引发情绪波动。由于编辑们并非匿名，不必要的严厉反馈最终会给你留下不好的印象。请使用与在会议上讨论研究或与导师面谈时类似的措辞。稿件同行评审是练习这些“软”技能的好方法，这些技能在科学界很重要，但经常被忽视。

幻灯片

成功标准

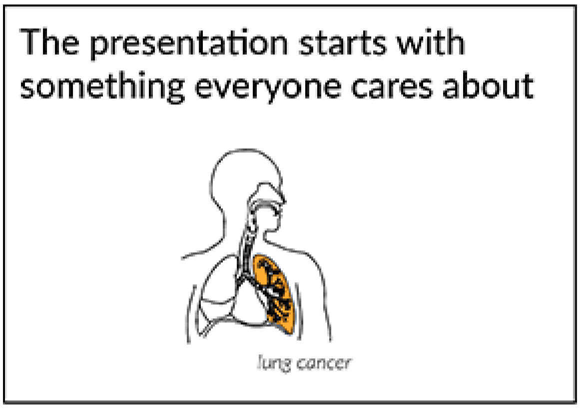
1. 演示从展示作品的更大动机开始。
2. 幻灯片中展示的研究具体地与更大的动机相关。
3. 实验及其结果与更大的动机相关。
4. 每张幻灯片都传达着一个信息。
5. 每张幻灯片提供的信息不会超过支持信息所需的范围。
6. 标题文本独立存在，大多数其他文本支持视觉效果。
7. 观众将会获得能够实现演讲者目标的信息。

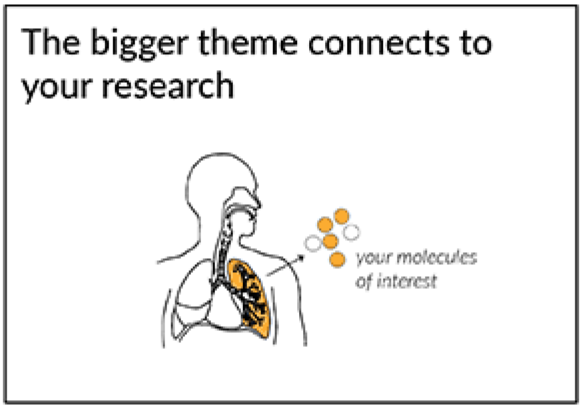
文档结构图

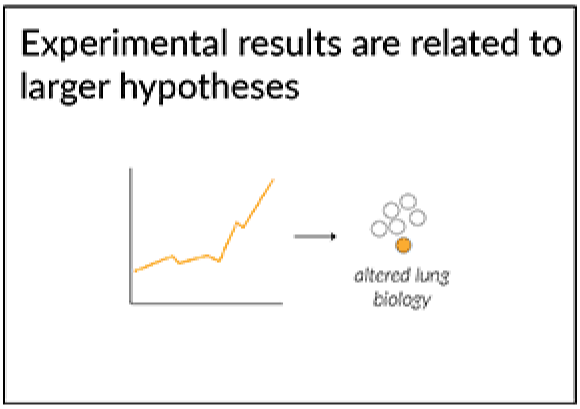
1. 我做这次演讲的目的是什么？
2. 我的听众是谁？
3. 成功的概念是什么？

- 将您的工作与更广泛的动机和假设联系起来
- “介绍”你的数据
- 每张幻灯片都应该传达一个观点
- 强调视觉效果而非文字
- 使每个图形尽可能简单，同时仍然传达其信息
- 避免使用行话、文字和视觉
- 为演讲的谈话部分做好准备

结构图







左上：报告应该从一个广泛关注的重要社会或科学问题切入，比如“肺癌”，以引起听众兴趣。

右上：讲述你自己的研究内容是如何与大家关心的问题（如肺癌）产生联系的，把研究嵌入到更大的科学图景中。

左下：不仅仅展示数据结果，还要说明它们如何支持或启发了更广泛的生物学机制或科学假设。

右下：幻灯片的最终目的，是为了帮助你实现演讲的目标，比如募资、合作、获得建设性建议等。

确定你的目的

了解自己作为演讲者的目标，并根据目标组织你的演讲。这样，听众在提供反馈时更容易理解你的逻辑，并找到最重要的要点。

例如，如果你的目标是从同事那里获得关于某个实验策略的反馈，那么就应该更多地关注实验方法。比较其与其他方案的优缺点。解释为什么你认为你提出的实验设计比其他实验设计能产生更有用的结果。

相反，如果你的目标是传达一个新的科学成果，那么你应该关注结果及其更广泛的含义，而不是你的方法论。避免谈论具体的方法（例如，说“我进行了基因表达”，而不是“我对一个 DSN 处理的转录组文库进行了测序”）。应该阐述你的发现如何影响更广泛的动机。

在不太正式的场合，比如实验室会议，你可以明确地告诉你的听众你在寻找什么（例如，“我希望对我的实验方法进行批评”）。

分析你的受众

不同的受众关注的点各不相同。工程师对创新感兴趣，科学家对新发现感兴趣，风险投资家希望了解盈利潜力，而临床医生则希望了解其如何帮助患者。您的演讲应该能够打动并激发受众的兴趣。

你可能比任何人都更了解你演讲的主题，所以你觉得自然而然的开场白，别人可能很容易感到困惑。你的演讲应该从大家都关心的话题开始，逐步阐述你实际做了什么以及为什么这么做。这看起来可能有点浪费。然而，如果你在背景资料上花费太多时间，你只会浪费几分钟。如果你在背景资料上花费的时间太少，你的听众可能无法理解你的工作，整个演讲就浪费了。

帮助观众理解演示文稿的另一种方法是避免使用行话，请记住，相同的单词或图像对一个观众来说可能是“有用的细节”，但对另一个观众来说可能是“行话”。

您的听众在听完您的演讲后会很高兴地了解到关键概念。

成功理念

将您的工作与更广泛的动机和假设联系起来

在演讲开始时，先概述你的工作背景，然后提出你想要回答的激励性问题。听众应该理解这些问题的答案将如何影响更广泛的背景。

在展示数据时，要清晰地阐明这些数据对这些激励性问题的启示。不同的实验结果对于这些更大的问题意味着什么？

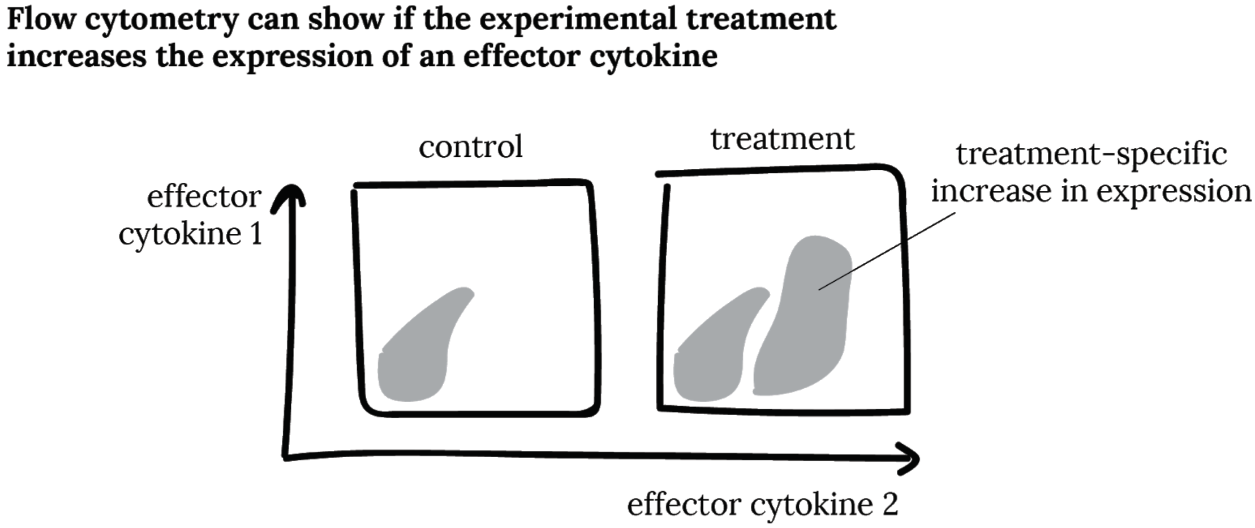
演讲各部分之间的过渡也应围绕更宏大的激励性问题展开。像“跑完这个凝胶电泳后，我决定也对 DNA 进行测序”这样不够有力的过渡，无法引导听众。更强有力的过渡可以是：“跑完这个凝胶电泳后，我们发现 PCR 产物丰富，但不确定这个条带是否来自我们想要研究的基因。之后，我们对 DNA 进行了测序，看看是否扩增出了正确的基因。” 更慢、更谨慎的过渡

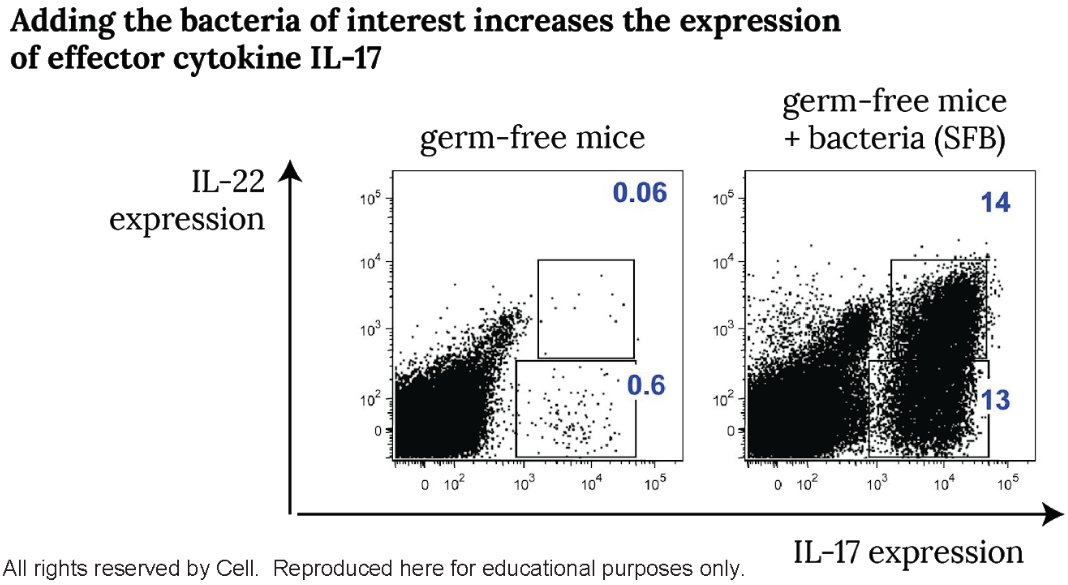
能让听众有时间消化他们刚刚看到的内容，并弄清楚如何将这些信息与演讲的其余部分联系起来。

“介绍”你的数据

在展示数据之前，确保你的受众能够理解。他们应该知道坐标轴是什么，图中的每个点代表什么，以及他们想要寻找的模式或信号。如果你向熟悉常见图表的受众展示这类图表（例如，给免疫学家看的 FACS 图，给合成生物学家看的基因回路图），就无需担心。但如果你在受众还不懂如何解读数据之前就展示数据，那么他们就会停止倾听，而是仔细查看图表，希望皱眉能帮助他们理解。

如果你担心观众无法理解你的数据，可以尝试用卡通画来展示假设成立时数据的样子，以及假设不成立时数据的样子。之后再展示真实的数据。





尝试以直观的方式解释不熟悉的数据。 对于不熟悉 FACS 图的观众，可以假设一位免疫学家先展示一张幻灯片，解释生物信号在 FACS 图中的呈现方式（上图），然后再展示包含实际数据的幻灯片（下图）。[改编自 Ivanov 等人，Cell (2009) doi:10.1016/j.cell.2009.09.033]

每张幻灯片都应该传达一个观点

保持信息简洁。每张幻灯片只讲一个要点。这样可以控制演讲的节奏和逻辑，并确保所有听众的理解一致。

幻灯片的标题应该是幻灯片的核心内容。换句话说，观众应该能够通过阅读幻灯片标题来理解你的故事。即使你完全忘记了幻灯片的内容，你也能通过阅读标题来了解幻灯片的大部分内容。

幻灯片类型	弱标题（只说形式）	强标题（传达信息）	原因说明
背景 slide	Antibiotics for Gram-negative pathogens	The antibiotic development pipeline is not limited by target discovery	强标题表达了研究动机
数据 slide	RT-qPCR	RNAi knocked down the target gene 100-fold	强标题直接呈现实验发现
结论 slide	Conclusions	真正的研究结论（如：Targeting X works）	强标题强调主结论，“结论”这个词已无需重复

把幻灯片标题变成你想让观众带走的关键信息，而不是描述你放了什么内容。

有力的标题能够传达信息。有力的标题往往是完整的句子，因为你需要一个完整的句子来传达信息。弱的标题往往是名词，例如“背景”或“细胞系的 RT-qPCR”。

这里有一种方法可以确定幻灯片的标题：让朋友看一下并说：“我不明白这张幻灯片的重点是什么。”如果你发现自己说：“好吧，我想通过这张幻灯片传达的是 X ”，那么 X 应该是幻灯片的标题。

如果幻灯片中有多个要点，请尝试以下操作之一：

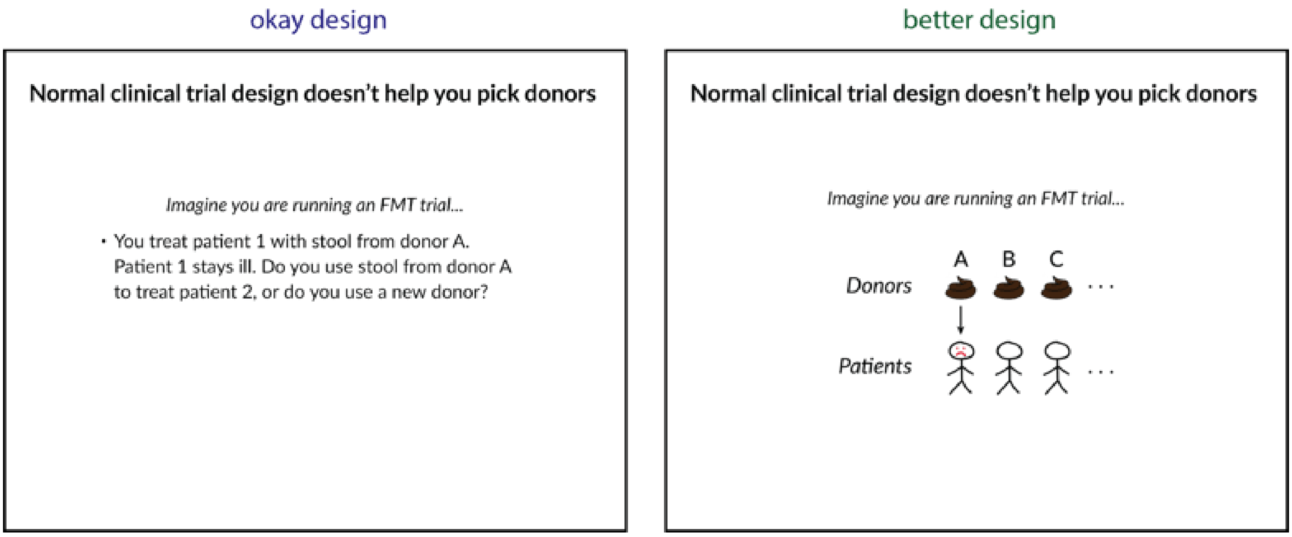
- 删除谈话中稍后不再提及的要点。
- 制作多张幻灯片，每张幻灯片都有自己的信息、标题和内容。
- 使幻灯片的各个部分出现和消失，以显示共同支持标题信息的不同内容。

强调视觉效果而非文字

当你放一张新幻灯片时，大多数观众会停止听你讲话，开始读幻灯片上的文字。大多数人根本听不清你在说什么。幻灯片上的文字越多，你对观众注意力的控制就越弱。如果你照着幻灯片念很多文字，你就失去了观众的注意力。

最好的情况下，一张幻灯片上只有一个完整的句子：标题。可以使用简洁的陈述来支持和强调标题尚未表达的次要结论。如果幻灯片上有大量文字，请尝试以下方法之一：

- 用图片替换文本
- 将幻灯片内容分成多张幻灯片，每张幻灯片阐述一个要点
- 把幻灯片上的文字取出来，放到演讲者笔记里



用易于阅读的图片代替大段文字。原始幻灯片（左图）在标题中阐明了要点，并用一个例子来佐证。该例子是一个长句。改进后的幻灯片（右图）传达了相同的信息，但用图片代替了文字。副标题（“想象一下你正在进行一项 FMT 试验”）是很好的辅助说明，而大段文字（“你治疗 1 号病人……”）则是供演示者阅读的脚本。

使每个图形尽可能简单，同时仍然传达其信息

图表的目的是通过视觉证据作为支撑来传达信息。你的受众通常会相信你，并认为你在图表中展示的内容对他们理解很重要。如果你展示的是详细的数据，你的受众就会被这些细节分散注意力，只见树木不见森林。

演示文稿中有效的图表通常不是直接用 Matlab 制作的，也不是你为论文制作的。与阅读学术论文或仔细研究自己的数据不同，你的听众没有太多时间仔细阅读图表。为了使图表更有效，请思考一下，图表至少需要展示多少内容才能阐明其观点。删除那些无助于证明观点的细节。简化数据标签，并使用颜色、箭头或标签来强调。

figure in a publication

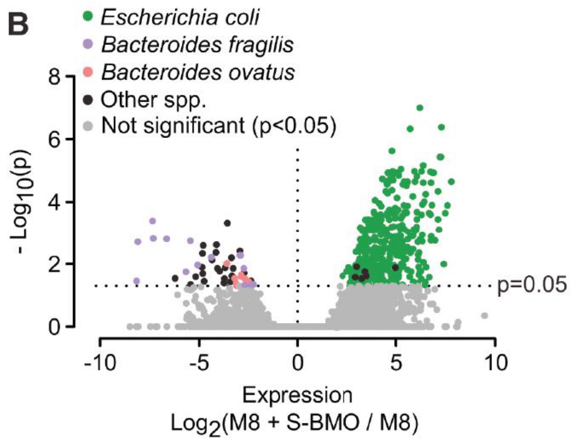
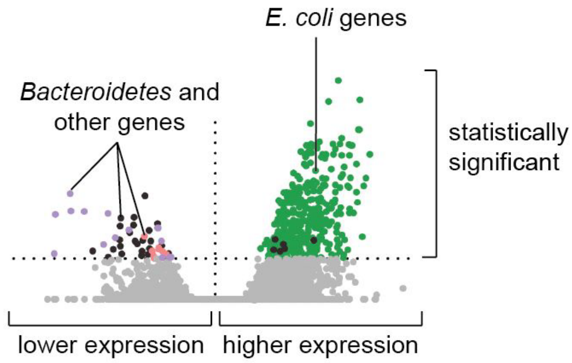


figure on a slide



简化标签并强调。论文中的图表（左图）非常详细：它显示了统计显著性、效应量以及计算效应量的方法。该图表的幻灯片版本（右图）省略了许多细节，但仍传达了要点：大肠杆菌基因表达上调，而其他物种的基因表达大多下调。原始图表的标题提到这是“细菌基因表达”；幻灯片版本通过在标签中直接使用“基因”一词来传达这一点。[改编自 Charbonneau 等人， Cell (2016) doi:10.1016/j.cell.2016.01.024]

figure in a publication

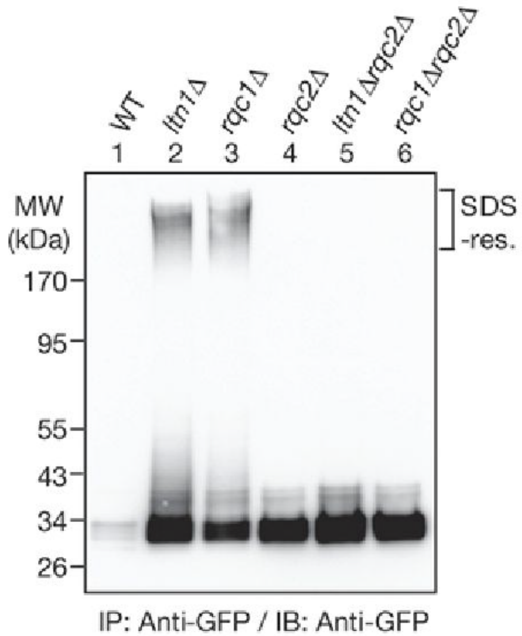


figure in a slide

genes knocked out	“non-stop” protein?	protein aggregate?
none	–	–
ltn1	✓	✓
rcq1	✓	✓
rcq2	✓	–
rcq2 and *ltn1*	✓	–
rcq2 and *rcq1*	✓	–

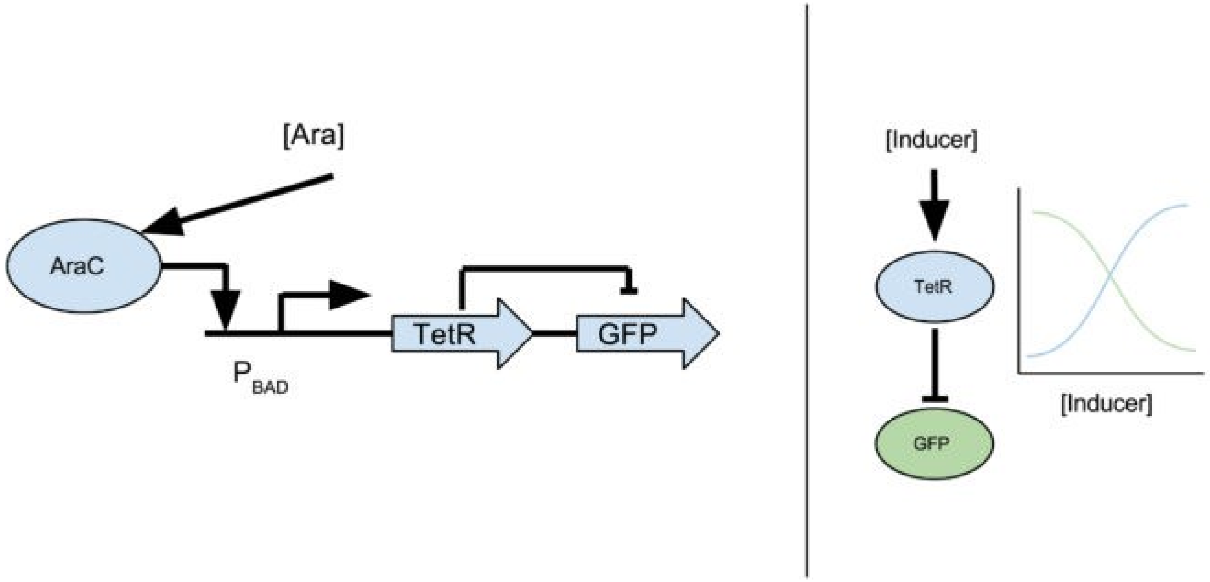
简化数据。学术审稿人、你的实验室同事和你的 PI 可能希望看到原始数据（左图），以

便他们自己进行解读。在演示中，你可能需要简化数据（右图），以强调实验结果（哪些基因敲除会导致哪些表型），而不是数据的具体形式（凝胶上的条带）。[改编自 Choe 等人，《自然》（2016）doi:10.1038/nature16973]

避免使用文字和视觉术语

帮助听众理解您的一个方法是避免使用专业术语。用听众都能理解的术语来讨论概念。特定基因或检测的名称可能广为人知，但它们可能会分散听众的注意力。例如，如果您说“GFP”，则不清楚是（a）任何旧的基因表达报告基因都能起作用，而您恰好使用了 GFP，还是（b）存在某些科学或工程原因，使得您选择 GFP 至关重要。

通常，“术语”指的是特定类型的词语，但一张图片胜过千言万语。如果你向观众展示某些细节——例如，你在实验中使用的基因回路的具体设计——那么他们会认为他们需要理解这些细节。试着将你需要理解的内容与观众需要理解的内容区分开来，前者是为了使实验成功，后者是为了理解你的结论。



避免使用视觉术语。（左）你可能在生物工程讲座中看到的基因回路图。诱导物（Ara）诱导转录因子（AraC）与启动子（PBAD）结合，进而激活另一个基因（TetR）的转录，而 TetR 则抑制 GFP 的表达。（右）简化的图表传达了重要的信息：增加诱导物浓度会增加 TetR 并减少 GFP。

为演讲的谈话部分做好准备

备演示文稿时，你很容易把所有时间都花在制作幻灯片上，但幻灯片只是演示文稿的视觉

辅助工具。演示文稿的重点是要有演示者！否则，你完全可以制作精美的幻灯片，打印出来，让观众阅读。

避免技术和程序问题，保持你的信誉并吸引观众的注意力。确保投影仪可以打开，电脑可以播放幻灯片，你最喜欢的幻灯片点击器已经准备好，幻灯片在灯光下看起来也不错等等。尽量在当天早些时候到达演示场地。想好你的站位，如何同步你的电脑和项目等等。

实际演示时，请站立不动，面向观众，并尽量减少手势。激光笔会分散注意力，而且难以使用。最好让幻灯片的重要部分自然突出。切勿背对观众。如果你的幻灯片简洁明了，背对观众应该会自然而然地做到这一点。记住：这次演讲是关于你和你的观众的，而不是关于幻灯片的。

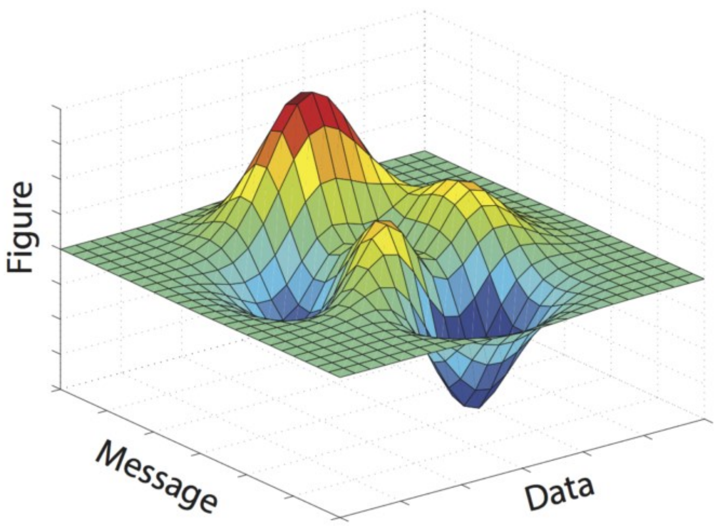
演讲前，做好问答环节的准备。弄清楚你可能会遇到哪些问题，并做好回答的准备。准备一些技术性更强的备用幻灯片，以解决特定问题。听到提问后，耐心等待提问者讲完。然后重复提问，并深吸一口气再回答。这个时间安排可以让其他听众理解问题，也给你一些时间来构思一个令人信服、条理清晰的答案。

图形设计

成功标准

1. 你的人物形象给观众留下了清晰的一句话的主要信息
2. 你提供直接支持主要信息的证据
3. 任何与你的主要信息无关的内容都会从图中删除

结构图



有效的数字是信息和数据的成功融合

目的

图表是对研究结果的任何视觉呈现，其形式多种多样。图表、示意图、照片、绘图、示意图和地图都属于图表类型。尽管形式多样，但所有图表的目的都相同——提供证据来支持您的信息。

你的意图就是你想向人们展示你的数据！一个好的图表不仅能向你的受众展示你的数据，还能传达你通过解读数据所获得的信息。

分析你的受众

精准的受众分析能让你更有说服力地传达信息。问问自己，谁会查看这个数据，他们是专家还是普通受众。根据答案，你需要调整信息及其支持证据的复杂程度。

例如，如果您的图表是发表在特定领域的期刊上，您的读者很可能对您的主题拥有丰富的背景知识。在这种情况下，您可以传达非常具体的信息（例如，“RTK 配体减弱致癌基因成瘾癌细胞系中的激酶抑制”来源：Wilson 等人，《Nature》，2012 年），并提供详细的证据（例如流式细胞术数据）。证据不足或过于简化会导致人们对您的信息产生怀疑。

如果您的图表将用于高中推广项目的演示，您的受众将拥有截然不同的背景知识。这时，您应该修改信息（例如，“癌细胞可能表现出对疗法的耐药性”），使其更易于理解。这可以通过提供更简单的证据（例如，免疫细胞的数量）来实现。向普通受众提供过多的细节会让他们不知所措；他们将无法从所有其他细节中区分出支持您主要信息的证据。

要点——向技术受众提供详细的证据以避免怀疑；向普通受众提供较少的细节以避免混淆。

最大化信噪比

将你想要传达的信息视为你的“信号”。你的目标是尽可能清晰地向受众传达这个信号。任何干扰你信息传达的因素都是“噪音”。我们已经讨论过如何通过优化图形设计来增强你的信号。这里我们将讨论如何最大限度地减少噪音。

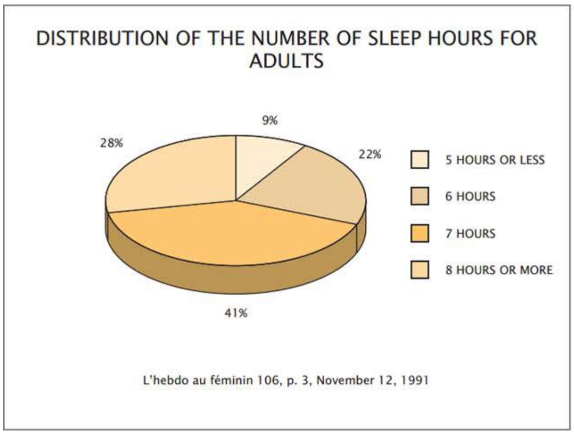
证据噪音

不要让听众被数据淹没：只包含阐明观点所需的最少数据。此外，包含不直接支持你观点的证据会分散听众对支持你观点的证据的注意力。

演示噪音

你呈现所选证据的方式也可能分散人们对你信息的注意力。以下图为例，我们将给出几个常见示例，说明如何改进图表以消除噪音。

- 图表的标题从数据的描述变为有关数据的信息。
- 图例直接移动到它们所描述的数据旁边，因此读者不必来回查看和匹配颜色。
- 配色方案经过简化和改变，以吸引人们对相关数据部分的注意。
- 删除了不必要的 3D 图形。



来源：Trees, Maps, and Theorems, by Jean-Luc Doumont, page 99

还有很多其他类型的干扰。例如，不必要的网格线或轴标签会使图表变得杂乱。问问自己，你希望读者从图表中获取什么，以及如何让他们更容易找到并关注相关信息。

（“信噪比”的比喻来自 Jean-luc Doumont’s book Trees, Maps, and Theorems.。）

MIT-电气工程与计算机科学系

图形设计

技能

选择最能传达您的信息的图形设计

正如不同的词语和句子结构在传达同一思想时效果可能不同，不同的插图设计在传达同一信息时效果也可能不同。在设计成功的插图时，要考虑哪些媒介、插图类型和情节类型最能凸显你的信息。

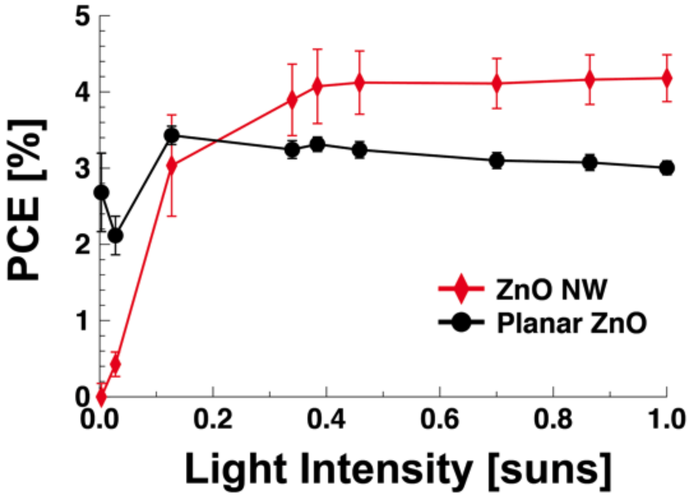
对于复杂的信息，多个面板可以将信息划分成清晰的陈述。多面板图表通常结合使用多种媒介、图表和情节类型。利用每个元素的互补优势来传达你的信息。

媒体类型可以以不同的方式传递相同的信息。

- 文字和语音表达精准。
- 表格呈现的信息很少带有背景或解释。
- 图表用证据说明了结论，并且可以进行解释。

举个例子，比较一下下面的表格和图表。虽然两者包含完全相同的数据，但图表却能提供一些解读，帮助读者识别趋势。

Light intensity [suns]	Efficiency with nanowires [%]	Efficiency without nanowires (planar) [%]
0.003	0	2.7
0.028	0.4	2.1
0.127	3	3.4
0.339	3.9	3.2
0.384	4.1	3.3
0.458	4.1	3.2
0.7	4.1	3.1
0.864	4.2	3.1
1	4.2	3



来源：Jean et al., “ZnO nanowire arrays for enhanced photocurrent in PbS quantum dot solar cells,” Advanced Materials 25, 2790–2796 (2013).

将这两种表述与“采用 ZnO 纳米线的胶体量子点太阳能电池比平面电池更高效”这一表述进行比较，该表述提供了一种在典型光照强度（约为 1 个太阳光）下有效的具体解释。阐明这一解释可以让读者清楚地理解，在充足阳光下的效率比在极低光照水平下的效率更重要。

曼彻斯特大学-学术短语库

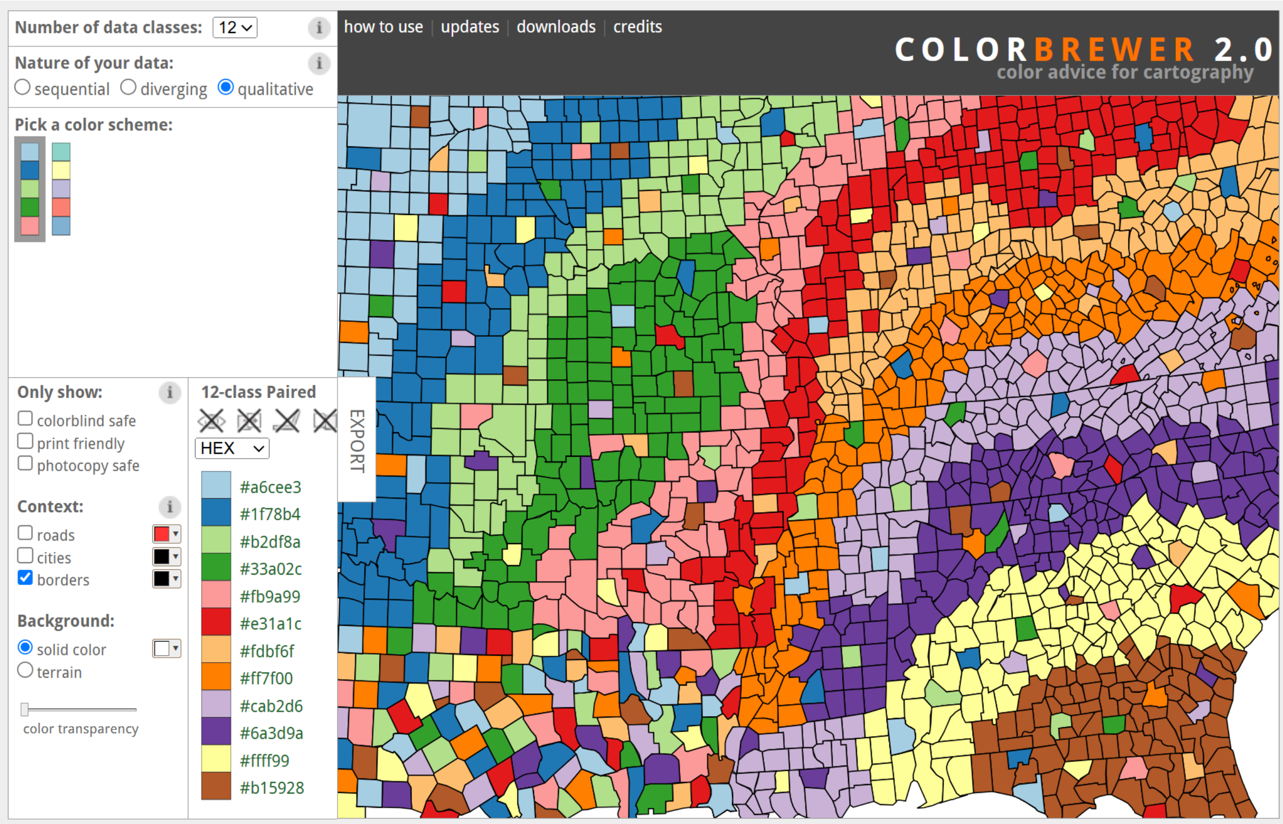
https://www.phrasebank.manchester.ac.uk/

学术短语库是面向学术写作者的通用资源。它旨在为您提供一些写作中常用的短语示例，这些短语根据研究论文或学位论文的主要章节进行组织。其他短语则列在学术写作中更常见的交流功能下。该资源对于需要报告研究工作的写作者尤其有用。这些短语及其标题可以简单地帮助您思考自己写作的内容和结构，或者您也可以在适当的地方将这些短语融入到您的写作中。在大多数情况下，使用短语时需要一定的创造力和适应性。学术短语库中的条目大多内容中立且具有通用性；因此，使用它们并非窃取他人的想法，也不构成抄袭。部分条目包含特定的实词，用于说明目的，使用这些短语时应替换这些实词。该资源主要面向非英语母语的学术和科学作家。然而，英语母语作家仍可能发现其中的许多内容很有帮助。事实上，最近的数据显示，大多数用户都是英语母语人士。

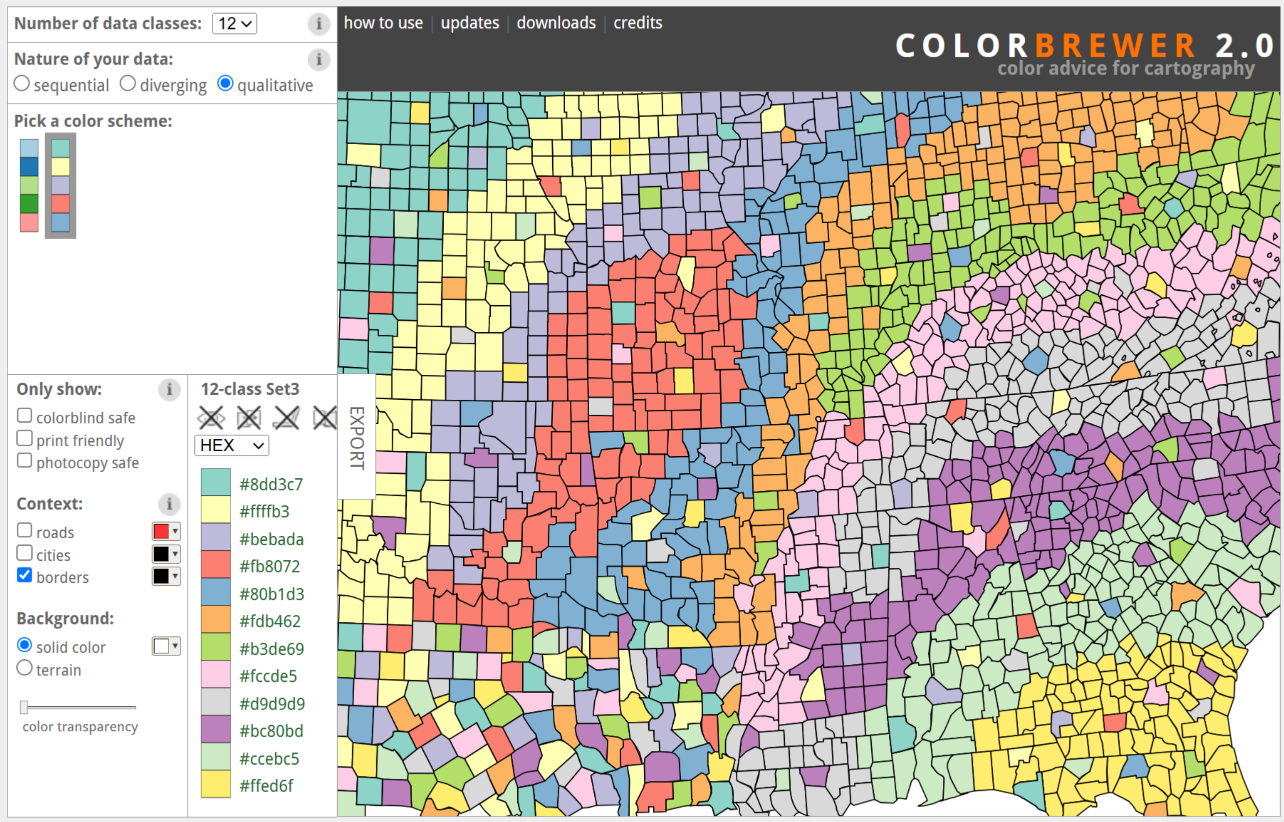
完整版可以参考《Academic Phrasebank 2018 Enhanced Edition》、《Academic Phrasebank 2023 Enhanced Edition》等。

配色

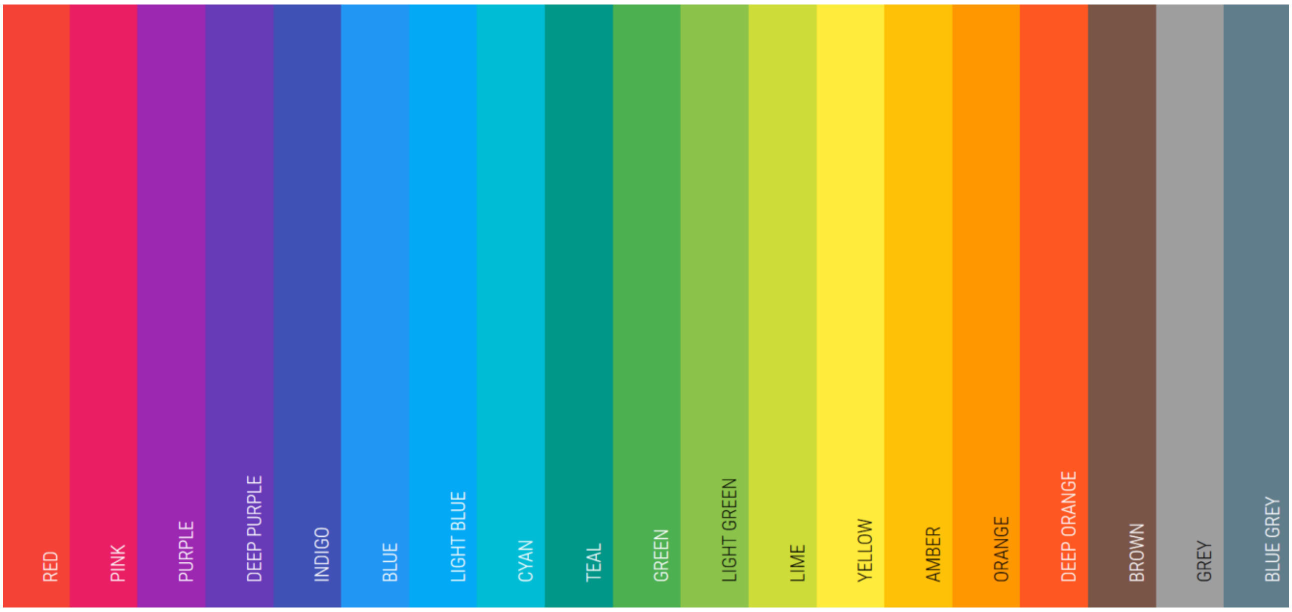
https://colorbrewer2.org/#type=qualitative&scheme=Paired&n=12



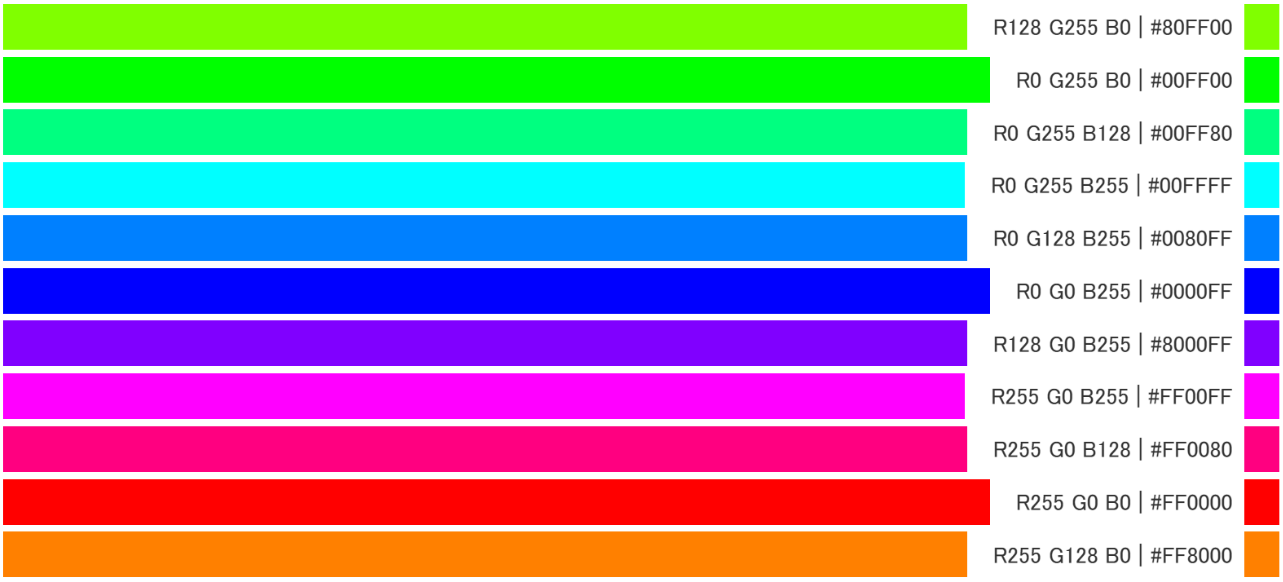
https://colorbrewer2.org/#type=qualitative&scheme=Set3&n=12



https://www.materialpalette.com/colors



https://ironodata.info/rgb.php?color=FDFF00



其他：

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- 《别告诉我你懂 PPT》

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后记

常听研究生谈及面对写作不知如何下手，所以我就综合了网上资源并简单翻译汇成这份教程，希望可以在期刊选择、论文写作、PPT 美化等方面提供一些参考。